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# How Closely Should You Follow a Trend? Atypicality and Engagement on Social Media

Marc Bravin is a Researcher and Lecturer at the Lucerne School of Computer Science, Lucerne University of Applied Sciences and Arts, Suurstoffi 1, 6348 Rotkreuz, Switzerland (email: marc.bravin@hslu.ch, phone: +41 41 228 24 58).

Melanie Clegg is an Assistant Professor at the Department of Marketing, Faculty of Business and Economics, University of Lausanne, Anthropole, 1015 Lausanne, Switzerland (email: melanie.clegg@unil.ch, phone: +41 21 692 34 49).

Reto Hofstetter is Full Professor of Executive Education and Marketing, University of St. Gallen, Dufourstrasse 50, 9000 St. Gallen, Switzerland (email: reto.hofstetter@unisg.ch, phone: +41 71 224 28 20).

Marc Pouly is Professor for Artificial Intelligence at the Lucerne School of Computer Science, Lucerne University of Applied Sciences and Arts, Suurstoffi 1, 6348 Rotkreuz, Switzerland (email: marc.pouly@hslu.ch, phone: +41 41 757 68 25).

Jonah Berger is Associate Professor of Marketing at the Wharton School, University of Pennsylvania, 3733 Spruce Street, Philadelphia, PA, USA (email: jberger@wharton.upenn.edu, phone: (215) 898-8249).

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# How Closely Should You Follow a Trend? Atypicality and Engagement on Social Media

## Abstract

Following trends on social media has become increasingly popular. But what is the best way to do so? Should brands and other creators copy the trend as closely as possible, or should they put a more unique spin on it? To answer this question, we develop a multimodal, unsupervised video analytics tool (MUVID) to quantify the typicality of over 85,000 TikTok dance videos. Results indicate that more atypical videos (i.e., more differentiated from the trend) generate more engagement. Consistent with the notion that atypicality drives engagement, this relationship is amplified when atypicality is easier to observe (i.e., when audiences have seen more trend videos). Follow-up experiments, including a content creator field experiment, manipulate atypicality and confirm its causal impact. Our findings provide practical guidance on how to create more impactful content, shed light on effective trend following, and offer a tool (available through an app) that researchers and practitioners can use to quantify typicality and analyze short videos more generally.

*Keywords:* trend following, atypicality, engagement, video analytics, social media.

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Videos, particularly short ones (typically between 15 and 60 seconds), have become the dominant form of social media content (Dencheva 2023).[1] The short-video platform TikTok reached one billion active users faster than any other social media platform (Dellatto 2021), and Instagram, LinkedIn, and YouTube soon jumped on the bandwagon, offering short, easy-to-digest video options (e.g., Reels and YouTube Shorts). As a result, to build and maintain their audiences, many brands and content creators are under pressure to generate up-to-date video content.

One way to address this challenge is to follow trends and imitate successful content patterns (Follet 2023). TikTok creators often follow dance trends, for example, performing a version of the same choreography to the same trending snippet of music (e.g., the famous choreography for the chorus of Gangnam Style released in 2012). Indeed, dances and dance challenges are among the top viewed categories on TikTok (ahead of other challenge types, Olsson 2023; Zabala 2025; Virós-Martín, Montaña-Blasco, and Jiménez-Morales 2024), "music" and "song" were the top two search terms on TikTok in 2025 (Cardillo 2026), and dances and songs are the first category shown on TikTok's "Explore" page. These trends also have a significant economic impact (Wang, Ramaprasad, and Gopal 2025).

Not surprisingly, then, brands and creators want to be a part of this phenomenon. Vogue, for example, followed the Macarena dance trend to present their spring collection (Satenstein 2021), and Ryanair posted dance videos of crew members who participate in trends (Influencer Marketing Hub 2022). Social media platforms (e.g., TikTok's "What's trending now") and ad agencies further encourage trend following by highlighting successful dance trends.

But when creating a video that builds off an existing trend, how should creators and

[1] Some consider videos up to 180 seconds as short videos (Tian, Dew, and Iyengar 2024), but we rely on the most typical format on TikTok (Metricool 2023).



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brands proceed? Should they imitate the existing trend as closely as possible (e.g., matching the core choreography), or should they deviate, creating their own, more unique variant?

When we asked content creators what they thought, over three-quarters of those surveyed considered trend following an important content strategy, but their opinions about the best way to do so were divided. While half (52%) thought closely following a trend was best, a sizable chunk (28%) thought deviating made more sense, and 20% did not indicate a preference (see Web Appendix A for details). Indeed, though prior research has examined other factors that encourage social media engagement, no research has examined the best approach for following existing trends.

A multimethod investigation, combining automated video analytics with controlled experiments, examines this topic. We start by developing a multimodal and unsupervised automated video analytics approach, MUVID, specifically tailored to quantify the similarity of short social media videos. Applying this approach to over 85,000 TikTok dance videos finds that more atypical videos receive higher engagement. Robustness checks, including analyses addressing potential confounds in the correlational data, and tests of trend exposure as a theory-relevant moderator, cast doubt on alternative explanations. Experiments, including a field experiment with content creators, directly test atypicality's causal impact and the role of trend exposure in driving these effects.

This research makes three key contributions. First, we contribute to the literature on social media engagement by examining the emerging phenomenon of social media trend following. Second, we contribute to research analyzing unstructured social media data (e.g., Berger, Humphreys, et al. 2020; Huang, Kaul, and Narayanan 2025; Feng et al. 2025; Kübler, Colicev, and Pauwels 2020) by developing MUVID as a multimodal unsupervised video

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analyzer for short video content analysis. Third, to make it easier for others to use MUVID for their own analyses, we provide an interactive online app publicly accessible via https://videoanalyzer.app/. The app allows users to (1) automatically extract and download audio and visual embeddings for short videos, (2) compute atypicality of videos, (3) predict content characteristics (e.g., category, quality, and suitable hashtags), and (4) predict engagement.

In what follows, we review related literature on social media engagement and discuss theoretical considerations. Then, we report the training procedure and architecture of MUVID and present evidence from secondary data, two online experiments, and one field experiment with content creators. We close with a description of the research app and directions for future work.

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## Social Media Engagement

Engagement is central to successful social media content. The more engagement content generates, the more likely it is to raise awareness, increase interest, and ultimately drive sales (Colicev et al. 2018; Lamberton and Stephen 2016). Not surprisingly, then, a large body of research has explored factors that make content more engaging. Beyond aspects of the content creator (e.g., their authenticity or number of followers; Beichert et al. 2024) or the context (e.g., time of day; Kanuri, Chen, and Sridhar 2018), a key component is obviously the nature and composition of the content itself (Dang et al. 2026).

Existing social media research has mostly focused on *text*, or the language used. Research on the drivers of sharing, for example, relied on text analyses to investigate the effect of arousal and public visibility (Berger and Milkman 2012; Berger and Schwartz 2011analyses of blog

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posts, tweets, and online conversations revealed reactions toward hedonic and utilitarian aspects, informative or emotional content, and sensory language (Borah et al. 2020; Büschken et al. 2025; Cascio Rizzo et al. 2023; Meire et al. 2019; see Berger, Humphreys, et al. 2020, and Packard and Berger 2024 for reviews).

As tools and content evolved, studies have begun to explore *images* as well (Liu, Dzyabura, and Mizik 2020; Feng et al. 2025). The mere presence of an image, for example, can increase engagement with a post (Li and Xie 2020). Similarly, pictures of consumers with products can boost engagement (Hartmann et al. 2021), calls-to-action can increase sharing (Villarroel Ordenes et al. 2018), and image composition, such as visual complexity, can influence engagement (Overgoor et al. 2022).

But despite the growing significance of *video* content, research on this content form is still relatively scarce (cf. Cheng et al. 2026; Tian, Dew, and Iyengar 2024; Wang, Bendle, and Pan 2024). Further, the limited research that exists has typically relied on manual coding or supervised learning to isolate specific video characteristics, such as slow-motion or facial expressions (Chakraborty et al. 2025; Lin, Yao, and Chen 2021; Stuppy, Landwehr, and McGraw 2024; Teixeira, Picard, and el Kaliouby 2014; Wals et al. 2026; Yang, Zhang, and Zhang 2025). Methods and tools for analyzing videos are still relatively scarce.

Moreover, no research has looked into the composition of videos created within a broader trend. Prior research on trends has either focused on characteristics of early trend adopters or whether trends can be detected early, rather than on how creators should follow existing trends. Lambrecht, Tucker, and Wiertz (2018), for example, show that early trend propagators are systematically less responsive to advertising, and other work developed methods to identify what may become trending topics in language (Jansen et al. 2021; Zhong and Schweidel 2020), but

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neither stream speaks to how content creators should participate in emerging trends.

Our research extends this literature by shifting the focus from trend emergence to trend following. To understand whether it is better to imitate or deviate from existing trends, we develop and present a novel approach to analyze videos and measure similarity in video content.

## Atypicality in Trending Videos

Videos that are part of larger dance trends vary in how much they adhere to the trend. Take the widely-covered "Savage Love" trend. Most clips center on one creator, waist-up in a vertical phone frame, swaying their elbows and hips to the "Savage Love" trend choreography in a neutral background. The camera stays static—propped on a dresser, locker, or outstretched arm (Figure 1, Panel A). Other clips, however, are more differentiated. They might include different choreography or unusual backgrounds (e.g., outdoors; see Figure 1, Panel B), special effects, multiple creators, eccentric outfits, or different camera angles.

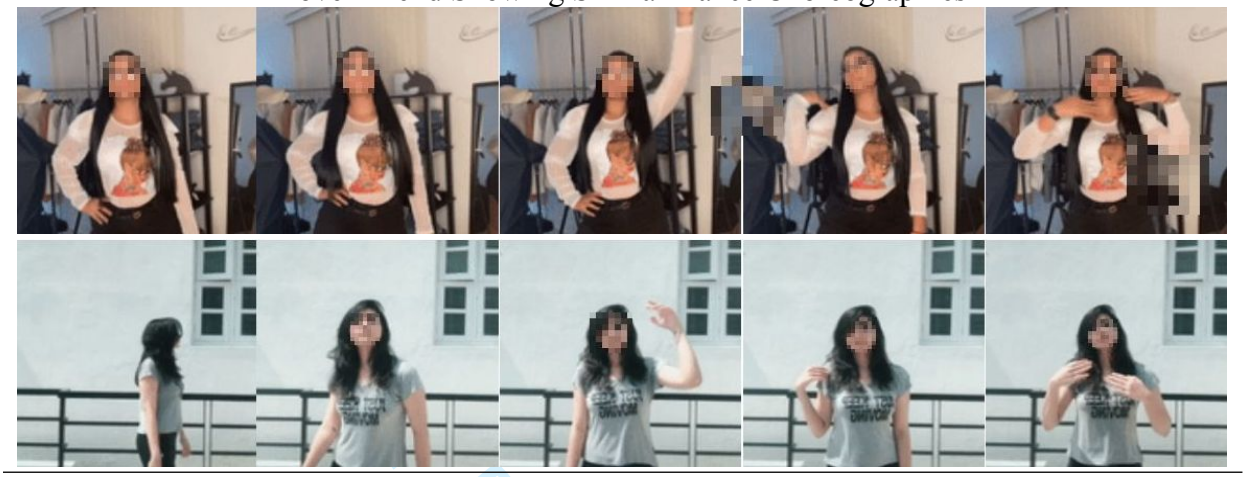
We start by asking how closely creators should copy an established dance trend. Existing theory offers competing predictions. Some work suggests that close adherence may facilitate engagement. Basic psychological research implies that exposure to a stimulus can increase the liking of similar stimuli (Monahan, Murphy, and Zajonc 2000), and work on media consumption shows that users often seek consistency in the content they consume (Schweidel and Moe 2016) or prefer consecutively watching content from the same category (Woolley and Sharif 2022). This might suggest that closely imitating a trend is most beneficial.

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Figure 1: Illustrative Examples of Typical and Atypical Interpretations of a Trend

Panel A – Typical Videos: Example of Two Creators Both Performing the "Savage Love" Trend Showing Similar Dance Choreographies



Panel B – Atypical Videos: Examples of Creators Showing Different Dances, Deviating in Background and Choreography



*Notes*: Panel A illustrates typical interpretations of the "Savage Love" trend, performed by creators against neutral backgrounds. Panel B presents atypical interpretations that diverge from the choreography, filmed outdoors in a more atypical background.

Other work, however, suggests that deviating may be better. Research on expectancy-violation, disfluency, habituation-saturation, and breaking plots, for example, suggests that repeated exposure to the same (or similar) stimuli can reduce interest (Burgoon 1993; Landwehr, Labroo, and Herrmann 2011; Wathieu 2004) and disconfirming built-up expectations can evoke interest and surprise (Labroo and Pocheptsova 2016). Indeed, novel and surprising things are more likely to be talked about and shared (Berger 2014; Berger and Iyengar 2013; Moldovan,

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Goldenberg, and Chattopadhyay 2011), and arousing content boosts social transmission (Berger 2011; Berger and Schwartz 2011; Cascio Rizzo et al. 2024). Consistent with this, firms garner more engagement when their content is distinguished from their closest competitors (Lysyakov et al. 2025) and songs do better on the Billboard chart when their lyrics are more atypical (Berger and Packard 2018).

Empirically addressing whether imitation or deviation is more effective requires being able to measure the similarities of videos (i.e., how similar one video's choreography and movement is to prior videos in a trend). For this purpose, we developed MUVID, a multimodal unsupervised video analyzer that transforms video files into numeric information (embeddings), which allows us to quantify a video's content for statistical analyses. Based on MUVID's embeddings, we first identify videos that are part of a particular trend. Then, we measure video atypicality, construct control variables, and test the link between atypicality and engagement.

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## MUVID: Extracting Video Embeddings to Measure Atypicality

Getting insights into video content is non-trivial (Grewal, Gupta, and Hamilton 2021) and prior marketing research has often relied on manual coding (e.g., Teixeira, Picard, and el Kaliouby 2014; Teixeira, Wedel, and Pieters 2010). This can be difficult to scale, however, and coding atypicality would be particularly challenging for human raters who would first have to learn several trends. Further, while automated video analysis can be more scalable and objective, existing work has primarily used supervised learning to detect specific features (e.g., people or products; Rajaram and Manchanda 2023; Schwenzow et al. 2021; Yang, Zhang, and Zhang 2025) or characteristics (e.g., facial expressions and gestures; Lin, Yao, and Chen 2021; Lu,

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Xiao, and Ding 2016; Wals et al. 2026) rather than measuring overall similarities. Consequently, neither approach can be directly transferred to measure atypicality, which requires tracking movements over time and analyzing the interplay between audio and visual elements.

Embeddings, however, can overcome the limitations of supervised feature extraction. These high-dimensional vectors numerically capture the essence of unstructured data and can measure similarity in text and images (e.g., Huang, Kaul, and Narayanan 2025; Shin et al. 2020). Recently, marketing researchers have begun extending this logic to video analysis. For example, Tian, Dew, and Iyengar (2024) utilized video embeddings to control for content characteristics when testing the effect of follower elasticity on engagement. Applying this approach at scale, though, presents a critical bottleneck: while computer science has developed models to extract video embeddings in an unsupervised manner (e.g., Qian et al. 2021), the models used in prior applied research are often proprietary or generalized. Indeed, to date, an open-source model specifically trained to capture the unique, fast-paced, and multimodal dynamics of social media short videos does not exist.

We apply the fundamental idea of unsupervised video analytics to extract video embeddings for social media short videos and name the resulting model MUVID (Multimodal Unsupervised VIDeo analyzer). MUVID can distill videos of variable durations into fixed-size embedding representations (i.e., vectors consisting of 1,024 numbers). This approach has several advantages. First, it is *multimodal*, incorporating synchronized audio and visual information. Second, it is *dynamic*, accounting for temporal aspects, movements, and object interactions. Third, it is *scalable*, as its unsupervised architecture does not rely on labeled data. Fourth, it is *versatile*, outputting embeddings that can be used for various downstream tasks. Finally, it is *open-source* and *accessible* through an interactive online app. Web Appendix B provides an

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overview and comparison of MUVID against existing video analytics models (Web Appendix Table W2 and section "Challenges of Automatic Video Analytics").

MUVID's embeddings are well-suited to detect videos belonging to a certain trend and to determine how similar a focal video is to others within that trend. We demonstrate this performance in a similarity-based retrieval task in Web Appendix B. Next, we provide details about the underlying sample and training process.

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***TikTok Video Dataset***

To train MUVID, we collected a large dataset of short videos from the most prominent short video social media platform, TikTok. TikTok videos usually combine video and audio elements, and TikTok's library offers licensed music, sound effects, and voice clips that users can incorporate into their videos. We sought a dataset of trending short videos from different creators and categories, and a sample that is representative of what users actually see on the platform, which is influenced by TikTok's feed algorithm (Klug et al. 2021). To achieve this, we applied a scraping strategy based on the three typical ways users normally access content on TikTok: through the feed, by using hashtags for video search, or by searching for a specific creator.[2]

We obtained a dataset of 864,294 videos from 126,845 different creators, uploaded between June 2015 and July 2022. Our dataset consists of creators with varying numbers of followers (min = 0, max = 144.8 M) and includes information about the videos (e.g., hashtags and music) and content-related statistics (e.g., number of views and likes). Web Appendix B ("Trend Video Datasets: Data Collection Process") provides more information about the data

[2] In line with typical video lengths on TikTok (Metricool 2023), we restricted our sample to videos that have a duration of 60 seconds or less, which constitutes more than 99% of all videos in our sample (median of 15 seconds).

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collection process and the resulting dataset.

***MUVID Model Architecture***

To account for video content's multimodal nature, we divide the model architecture into two neural networks, a visual network and an audio network (see Figure 2). The visual network consists of an R(2 + 1)D ResNet-18 model (Tran et al. 2018), which is fed with short clips consisting of 16 frames of dimension 112 × 112 pixels, and transforms them into a 512-dimensional embedding vector. In comparison to 2D CNNs, which are often used for images or audio, this 3D CNN enables the model to capture entangled spatial and temporal information within the video clips and extract meaningful features from the video data, such as motion patterns and object interactions.

The audio network processes raw audio signals by first converting them into Mel spectrograms, which represent the frequency content of the audio over time (Al-Tahan and Mohsenzadeh 2021). A 2D ResNet-18 (He et al. 2016) is then applied to these spectrograms to generate an additional 512-dimensional embedding vector. This 2D CNN captures temporal information by applying convolutional filters across both time and frequency dimensions, enabling the learning of local temporal patterns (Hershey et al. 2017). Pooling layers further retain essential features while ensuring invariance to minor temporal shifts, effectively encapsulating the audio's frequency and temporal characteristics.

To train MUVID, we use an unsupervised training procedure that does not rely on manual video labeling. In Web Appendix B, we explain our training procedure in depth and report several quantitative and qualitative indicators for the model's performance, including several downstream evaluation tasks we performed as suggested by prior literature (Schiappa, Rawat,

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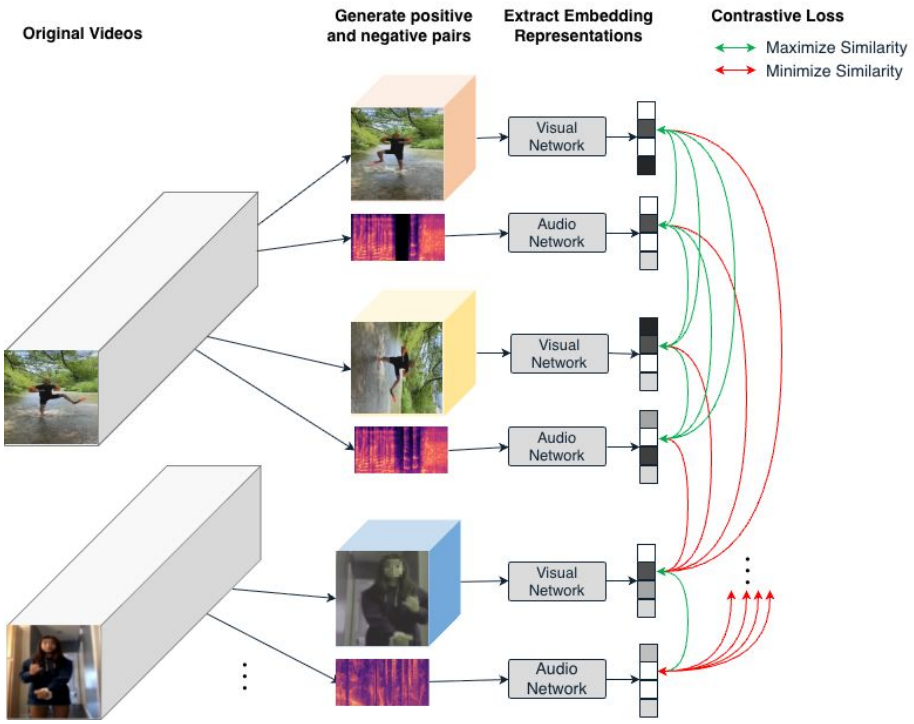
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and Shah 2023).

We leverage the extracted MUVID embeddings to answer our substantive research question, and provide the extracted embeddings and use cases in a click-through online web converter app (discussed in more detail before the General Discussion). Crucially for our empirical application, MUVID yields distinct 512-dimensional audio and visual embedding spaces rather than a single fused vector (labeled as visual_embedding1-512 and audio_embedding1-512 in the app). This allows the modalities to be utilized independently depending on the analytical task. For example, the audio embeddings can be isolated to identify trends (which are defined by sound), while the visual embeddings can be used to measure visual deviations.

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Figure 2: Overview of the MUVID Architecture and Pre-Training Procedure




*Notes:* We created two augmented versions of both the video frames and the audio signals for each video, which are then fed into a visual neural network and an audio neural network, respectively, to generate embedding vectors. We trained the model by maximizing the similarity between representations drawn from the same video while minimizing the similarity between representations from different videos. In this way, visual and audio information are semantically entangled, and no labeled data are required for the training procedure.

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Based on the MUVID embeddings, we next present model-free and model-based evidence testing the relationship between atypicality and engagement, as well as several robustness checks. Then, we use experiments to test the causal nature of the relationship documented using MUVID, and further test how users should deviate. Table 1 provides an overview of our empirical package.

Table 1: Overview of Studies

| | Overview | Design | Key Finding | Contribution |
|---|---|---|---|---|
| **Study 1** | Analyses of 85,000+ TikTok videos, multiple trends | Grouped logit with creator-, trend-, and time-fixed effects | Atypicality positively associated with engagement ($\beta = .025$, $p < .001$); effect stronger for deviations occurring later within the video and later in the trend's lifecycle | Observational evidence for main effect; identifies trend exposure and deviation timing within a video as moderators |
| **Study 2** | Online experiment | 2-cell between-subjects experiment (atypical vs. typical) | Atypicality causally increases engagement (24% vs. 13%) | Provides causal evidence for main effect |
| **Study 3** | Online experiment | 2 × 2 between-subjects experiment (atypicality × trend exposure) | Atypicality increases engagement under high trend exposure (34% vs. 18%); effect absent under low exposure | Provides causal evidence for moderation via trend exposure |
| **Study 4** | Field experiment with TikTok creators | Within-creator experiment (atypical vs. typical) | Atypicality increases engagement in the field (unstandardized b = .093, p = .044; ~10% higher odds of a like) | Demonstrates external validity for causality of the main effect in a real-world TikTok setting |

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## Study 1: Video Atypicality within TikTok Trends

Our first study uses MUVID's generated video embeddings to explore whether more atypical videos receive more engagement. The embeddings are used to (a) identify trends, (b) assign videos to the trend, and (c) determine the atypicality of each video to its trend.

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***Trend Dataset***

First, we constructed a trend dataset. Given that dancing trends are one of the most-watched categories on TikTok (Ceci 2022), we focus on trending dance videos.

*Trend identification*

When following trends, creators use the same part (i.e., music snippet) of a song, which is usually 15-30 seconds long and typically identifies a trend. To find videos that belong to a dance trend, we first select dance videos by training a binary classifier that distinguishes dance from non-dance videos, using transfer learning on MUVID's visual embeddings. As a next pre-processing step, we make use of the MUVID audio embeddings to assign dance videos to trends. Since direct indicators of which trend a dance video belongs to are lacking[3], we apply the DBSCAN clustering algorithm (Ester et al. 1996) to group videos using the same music snippet according to the MUVID audio embeddings, accounting for song title inaccuracies or different segments of the same song. We keep clusters with at least 10 videos, resulting in 86,244 videos grouped into 1,548 trends.

*Atypicality calculation*

To quantify the atypicality of videos within a trend, we develop an objective measure using the MUVID visual embeddings. We adopt a distance-based approach, a robust and widely used paradigm in unsupervised anomaly detection (Aggarwal 2017). The core assumption of this paradigm is that normal data points lie in dense neighborhoods, while anomalies are located far from their neighbors. This paradigm is particularly synergistic with our MUVID model because the underlying contrastive learning objective is designed to create exactly this type of geometric

[3] TikTok provides music labels, but these labels are oftentimes incorrect and they do not provide information about the specific part of a song that is used within a video.



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structure in the embedding space. Specifically, the contrastive loss optimizes for two key properties: alignment, which pulls semantically similar videos into tight, dense clusters, and uniformity, which ensures that dissimilar videos are spread out across the feature space (Wang and Isola 2020). The resulting structure, where typical videos are clustered and atypical ones are isolated in sparse regions, makes distance-based scores a direct measure of atypicality. We do not use the audio information, as it is fixed for all videos within a given trend.

Importantly, we compare focal videos $\boldsymbol{v}_i$ to earlier videos submitted to the same trend. We calculate the average of the distances between the focal video $\boldsymbol{v}_i$ and all *previous* videos within the same trend, a technique conceptually similar to k-nearest neighbor (k-NN) based anomaly scoring (Angiulli and Pizzuti 2002). This results in the following equation:

$$atypicality_i = \frac{1}{|\eta(t_i)|} \sum_{v \in \eta(t_i)} dist(\boldsymbol{v}_i, \boldsymbol{v}) \quad (1)$$

where $\eta(t_i) = \{ \boldsymbol{v}_j : t_j < t_i \}$ is the set of videos from the same trend that were posted before the focal video *i*, and $dist$ is a function that computes the distance between two video embeddings.

The cosine distance is used to measure dissimilarity between videos' embeddings, and we average the distance to all previously posted videos in the same trend (similar to Lysyakov et al. 2025; Wei, Hong, and Tellis 2022; Yang, Zhang, and Kannan 2022). The greater the distance, the more dissimilar the video is to all other videos in the trend. Hence, higher scores indicate that videos are more atypical.

*Engagement measure*

Higher engagement indicates that more users are actively interacting with a post (Colicev et al. 2018). Following prior work (e.g., Rajaram and Manchanda 2023), we focus on likes and take the number of views into account, capturing engagement conditional on viewing (i.e.,



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proportion of viewers who like a video). Formally, this engagement rate can be expressed as:

$$engagement_i = \frac{likes_i}{views_i} \quad (2)$$

Though this measure cannot fully rule out targeting or audience-selection effects, it mitigates potential confounding effects from platform algorithms that affect exposure (Klug et al. 2021). Substantively, the measure captures how appealing a video is once it has been seen. For example, a value of .10 indicates that, on average, 10% of viewers who saw a video also liked it.

Given these atypicality and engagement calculations, we construct a cross-sectional dataset with the video as the unit of analysis. That is, each observation is a video, produced by a creator and assigned to a specific trend, and each creator can have videos in multiple trends. Atypicality is measured at the time of posting, in comparison to all previous videos, while engagement is measured at the time of data collection. All sample details are provided in Web Appendix C ("TikTok Trend Dataset").

The empirical analyses model engagement rate using a grouped logistic regression that treats likes as successes out of the total number of views.

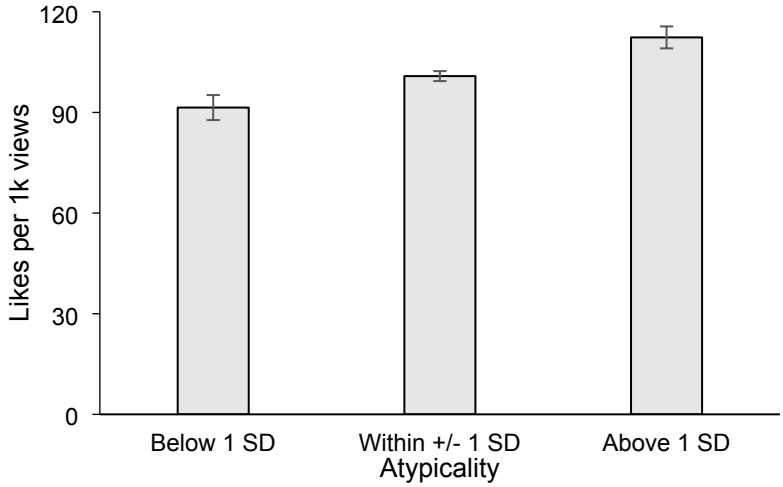
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***Model-Free Evidence***

We first descriptively examine the relationship between atypicality and engagement. Model-free evidence suggests that atypical TikTok trend videos receive more engagement (Figure 3): Highly atypical videos (> +1SD above average) receive more engagement (143.6 likes per 1,000 views) than videos with average atypicality (between -1 and +1SD; 127.7 likes per 1,000 views), which receive more than videos that are not very atypical (< -1SD; 111.9 likes per 1,000 views).

Figure 3: Model-Free Evidence for the Relationship Between Engagement and Atypicality

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*Notes:* The average number of likes per 1,000 views for videos with different levels of atypicality. Error bars = ± 1 SE.

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***Modeling Approach***

The goal of our modeling approach is to test the relationship between atypicality (detected with MUVID embeddings) and engagement. To do so, we estimate a grouped logit using a generalized linear model (GLM) with a binomial family and logit link, modeling the number of likes as the number of successes out of the total number of views (i.e., trials, Agresti 2013). The model is "grouped" because likes and views are observed only as aggregated measures at the level of individual videos (i.e., total number of likes and the total number of views).

Compared with modeling the like rate directly using standard linear regression, this approach naturally incorporates the varying reach (i.e., number of views) of different videos by weighting observations by the number of trials. Further, it ensures that predicted engagement remains within the valid 0-1 probability range. As is standard in logistic regression, coefficients are estimated in log-odds and reflect the association between atypicality and a viewer's likelihood of liking a video. Notably, our results remain consistent when estimating an Ordinary

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Least Squares (OLS) or Poisson model (see Web Appendix C), supporting the robustness of our findings.

To account for some potential confounds and examine the association between atypicality and engagement, we estimate a comprehensive model that controls for video-, creator-, and trend-specific characteristics.

First, we control for video-level confounders. Although most engagement typically occurs shortly after posting (Graffius 2025), results could be influenced by how long a video has been online. Perhaps atypical videos garner more engagement simply because they accumulated likes over a longer period. Accordingly, we include Time on Platform (the number of days from posting to data collection) as a control. Alternatively, atypical videos may inherently have higher visual quality or involve better dances, and such factors, rather than atypicality, might drive engagement. To address this possibility, we use the MUVID embeddings to predict dance quality and visual quality (see Web Appendix C, "Quality Control Variables" for details) and include these indicators as further controls.

Second, we account for creator-level heterogeneity using a combination of time-varying controls and fixed effects. Aspects of the video creators heavily influence both atypicality and engagement. Because we collected data at multiple time points, we are able to include time-varying, observable controls for a creator's evolving status and activity. We control for Creator Followers and Creator Verified Status, as higher-status creators tend to generate more engagement (Wies, Bleier, and Edeling 2023) and might produce more atypical content. We also control for Creator Productivity (total number of videos posted; Shriver, Nair, and Hofstetter 2013), which affects overall engagement and content variance. Beyond these observable metrics, we include creator-fixed effects to hold constant all unobserved, time-invariant characteristics of

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individual creators, such as innate dancing talent, personal aesthetic, or baseline audience loyalty. This ensures that any link between atypicality and engagement relates to the content of the video itself rather than the underlying idiosyncrasies of the person posting it.[4]

Finally, we include trend- and time-fixed effects to account for broader environmental contexts. Trend-fixed effects control for the possibility that certain trends inherently allow for greater variation in video interpretation or garner different baseline levels of engagement. Time-fixed effects (monthly) account for platform-level fluctuations, such as the surge in TikTok's popularity during the COVID-19 pandemic (Dixon 2022). The final model equation is given as follows. Let $like_i$ denote the number of likes and $view_i$ the number of views for a video $i$ of a creator $j$. We model likes as binomial outcomes with $view_i$ trials and success probability $p_i$. Specifically:

$$like_i \sim Binomial(view_i, p_i)$$

and

$$logit(p_i) = \beta_0 + \beta_1 \text{Atypicality}_\text{i} + \sum_{k=1}^{K} \beta_k \text{Control}_{kij} + \text{TrendFE}_t + \text{CreatorFE}_j + \text{TimeFE}_m \quad (3)$$

Where $p_i$ denotes the probability that a view results in a like. $logit(p) = log\left(\frac{p}{1-p}\right)$ is the logit link function; $\beta_0$ is the intercept; $\beta_1$ is the coefficient for atypicality, representing its association with the log-odds of engagement; $\beta_k$ are coefficients for the $K$ other control variables; TrendFE (trend-fixed effects) account for variations in engagement across different

[4] Reverse causality could also be a concern (e.g., creators who gain popularity might systematically shift toward more atypical content as their confidence or audience tolerance grows). We test for this possibility in our robustness checks by examining whether a creator's atypicality changes systematically over their chronological posting order, and find no evidence of such a pattern (see section "Robustness Checks").



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trends; CreatorFE (creator-fixed effects) control for unobserved, time-invariant characteristics of the individual creators, such as innate talent or baseline aesthetic; TimeFE (time-fixed effects) account for changes in engagement over time; Standard errors are clustered by trends. We standardize all continuous variables to ensure comparability of scales.

***Assumptions and Empirical Challenges***

Our modeling approach makes several assumptions. First, we assume users must have been previously exposed to a trend to detect atypicality. To test this assumption, we surveyed TikTok users and found that the most prevalent trends are known by up to 80% of users (details in Web Appendix D). Second, we assume users engaged with these videos shortly after posting, because this is when we measure atypicality. Supporting this assumption, 80% of all comments in our dataset were accumulated in a median of four days after posting the video. Prior research and industry sources corroborate this, indicating that most social media content garners engagement shortly after posting, and on TikTok, most engagement can even emerge within a couple of minutes or hours (Hartmann et al. 2021; Graffius 2025). Thus, it is plausible to assume that users are familiar with trends, particularly as trends become more popular, and that engagement usually accumulates shortly after posting. Sample statistics and an overview of included variables are provided in Web Appendix Table W9. Web Appendix Table W12 provides a comprehensive overview of empirical challenges in our modeling approach and how we addressed them.

***Results***

Even controlling for video quality, creator characteristics, and creator-, trend-, and time-

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fixed effects, results indicate a positive and significant relationship between atypicality and engagement ($\beta = .025$; $SE = .007$; $t = 3.65$; $p < .001$; Table 2, Model 1). Specifically, a one-standard-deviation increase in atypicality is associated with a 2.5% increase in the odds that a view results in a like (Odds Ratio = 1.025).

While a 2.5% increase may appear modest, its practical value is substantial when scaled to the high-velocity environment of TikTok. To illustrate the economic magnitude of this relationship, we compare the expected outcome of a "typical" video (one standard deviation below the mean in atypicality) versus an "atypical" video (one standard deviation above the mean). Using the mean engagement rate of approximately 10% (which reflects the baseline average of our sample), this two-standard-deviation shift corresponds to an increase of roughly 0.46 percentage points in the probability that any given view results in a like. For a median creator in our sample, characterized by approximately 50,000 followers and averaging 15,000 views per video, this translates to a projected increase of approximately 69 organic likes per single video. Over a year of consistent posting—for example, three videos per week, which aligns with industry recommendations for posting frequencies (Lang 2025)—this accumulates to roughly 10,800 additional organic likes, meaningfully boosting channel growth.

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Table 2: Relationship Between Video Atypicality and Engagement

| DV | Engagement | | |
|---|---|---|---|
| Model | (1) | (2) | (3) |
| **Atypicality** | **.025*****<br>*SE* = .007, *p* < .001 | **.018****<br>*SE* = .009, *p* = .048 | |
| Time on Platform | **-.003**<br>*SE* = .031, *p* = .916 | **-.025****<br>*SE* = .012, *p* = .038 | **-.004**<br>*SE* = .032, *p* = .892 |
| Dance Quality | **-.010**<br>*SE* = .009, *p* = .249 | **-.014**<br>*SE* = .010, *p* = .148 | **-.010**<br>*SE* = .009, *p* = .277 |
| Visual Quality | **-.012**<br>*SE* = .007, *p* = .093 | **-.015***<br>*SE* = .008, *p* = .064 | **-.012**<br>*SE* = .007, *p* = .099 |
| Creator Followers | **.003**<br>*SE* = .004, *p* = .558 | **.005**<br>*SE* = .005, *p* = .347 | **.003**<br>*SE* = .005, *p* = .497 |
| Creator Verified Status | **-.0001**<br>*SE* = .018, *p* = .995 | **-.050*****<br>*SE* = .017, *p* = .003 | **-.001**<br>*SE* = .018, *p* = .977 |
| Creator Productivity | **-.010**<br>*SE* = .027, *p* = .703 | **.011**<br>*SE* = .023, *p* = .624 | **-.011**<br>*SE* = .027, *p* = .696 |
| Trend Exposure | | **-.038*****<br>*SE* = .010, *p* < .001 | |
| **Atypicality × Trend Exposure** | | **.029****<br>*SE* = .009, *p* = .001 | |
| **Begin Atypicality** | | | **.005**<br>*SE* = .010, *p* = .607 |
| **End Atypicality** | | | **.025****<br>*SE* = .011, *p* = .023 |
| Trend-fixed Effects | Included | | Included |
| Creator-fixed Effects | Included | Included | Included |
| Time-fixed Effects | Included | Included | Included |
| Observations | 74,227 | 72,925 | 74,227 |
| AIC | 193,825,955 | 268,813,944 | 193,515,498 |
| BIC | 193,952,568 | 268,925,818 | 193,642,119 |

*Notes:* All continuous variables were z-normalized to ensure that they are on similar scales. Unstandardized results do not differ in sign or significance. For Model 2, we removed trend-fixed effects because trend exposure is constant within a trend. DV = dependent variable; SE = standard error; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion. * $p < .05$, ** $p < .01$, *** $p < .001$.

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The impact becomes even more pronounced for top-tier creators. For a macro-influencer averaging 1,000,000 views per video, the same shift toward atypicality translates to roughly 4,600 additional likes per video. In the influencer marketing industry, where engagement is a primary currency, this increase holds direct financial value. Industry benchmarks estimate the Cost Per Engagement on TikTok at $0.27 (Aspire 2024). Applying this benchmark, these 4,600 additional likes represent an added media value of over $1,200 per single video. This demonstrates that for professional creators, optimizing for atypicality is not merely a stylistic

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choice but a substantial lever for increasing the financial value of their content.

***Robustness Checks***

Although we included multiple relevant controls, unobserved variables may influence both atypicality and engagement. For instance, content creators may systematically differ on unobservable dimensions (e.g., strategic intent), which might influence both their likelihood of posting atypical content and the subsequent engagement their content receives. Accordingly, we ran a series of robustness checks (see Web Appendix C), including:

1) Propensity score matching to account for potential self-selection (confirming the relationship holds when each atypical video is compared to a typical video matched on observable characteristics such as video quality and creator attributes).
2) A Gaussian copula approach to account for unobserved heterogeneity in creators (finding no evidence of endogeneity biasing our estimates).
3) Testing for creator learning by checking whether atypicality changes over a creator's posting sequence, to rule out reverse causality (finding no systematic change in a creator's atypicality over time).
4) Ruling out shifts in the viewer population via linguistic style matching of user comments (confirming that the underlying audience remains stable across both typical and atypical videos).
5) Controlling for video text elements to ensure written information is not driving the effects (showing the main effect remains positive and highly significant).
6) Verifying measurement robustness through alternative atypicality operationalizations and trend-size thresholds (yielding consistently positive and significant effects across

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configurations).

7) Testing alternative model specifications via Poisson and OLS regressions (both confirming the positive, significant effect of atypicality on engagement).

Across all of these robustness checks, our core finding remains unchanged: following a trend with higher atypicality consistently remains positively correlated with engagement.

***Testing the Role of Trend Exposure***

While the results so far are consistent with the notion that atypicality positively affects engagement, one could wonder if other confounding factors may explain the observed relationship. To further probe this possibility, we examine trend exposure as a potential moderator. To see something as atypical, it helps to have a sense of what typical is in the first place. If the first bird children saw was a flamingo, for example, they would not necessarily think it was an atypical bird. It is only through repeatedly seeing robins, finches, and other more "typical" birds that one comes to realize a flamingo is rather unusual.

Consequently, if atypicality is driving the effects, then the relationship should be moderated by the degree to which the audience has already been exposed to the trend. The more videos of a given trend have been posted so far, the more likely viewers have been exposed to trend videos previously, which should make it easier for them to judge whether a new focal video is atypical. Indeed, survey evidence confirms that trend exposure predicts how familiar users are with a trend ($\beta = .48$, $p < .001$; details in Web Appendix D). Consequently, we test whether trend exposure (i.e., the total number of videos posted so far within a trend) moderates the effect. We use the same model as in Study 1, including the same control variables and time-fixed effects for this analysis, but removing trend-fixed effects because trend exposure is

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constant within a trend.

Consistent with our suggestion, results indicate that trend exposure moderates the link between atypicality and engagement ($\beta = .029$; $SE = .009$; $t = 3.42$; $p = .001$; Table 2, Model 2). The more videos that are already out there for a particular trend, the stronger the link between atypicality and engagement. While later experiments (Studies 2, 3, and 4) directly test causality, this finding is consistent with the notion that atypicality may be increasing engagement.

***Discussion***

Both model-free and model-based evidence indicate that atypicality is positively associated with engagement. This effect remains robust across different modeling strategies and controlling for video quality and time-varying creator characteristics. Importantly, the effect holds within creators (via creator-fixed effects), suggesting that deviating from a trend's typical form increases engagement even when creator identity, talent, and style are held constant. Further, consistent with the notion that atypicality is driving this effect, moderation analysis indicates the link between atypicality and engagement is strongest when audiences are more likely to have seen the trend already.

Ancillary analyses also find that, consistent with the notion that audiences need to recognize the trend, atypicality matters more at the end of videos rather than the beginning. People can only appreciate novelty when they can make sense of "surprising" elements and deviations (Elsen, Pieters, and Wedel 2016). Yet for this "surprising" element to occur, patterns must be recognized before deviations can evoke interest (Loewenstein, Raghunathan, and Heath 2011). This suggests that users must first recognize the trend prototype before they can appreciate a deviation from it, and consequently, ending atypically may be more beneficial.

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Splitting the trend videos into two parts (the initial and final 5 seconds of the video) and separately estimating the atypicality for each part using MUVID's visual embeddings finds evidence consistent with this notion. While deviations in the final part of the video have a positive and significant correlation with engagement ($\beta$ = .025; $SE$ = .011; $t$ = 2.27; $p$ = .023; Table 2, Model 3), this disappears for atypicality in the initial part of the video ($\beta$ = .005; $SE$ = .010; $t$ = .51; $p$ = .607; Table 2, Model 3).

Overall, while results of Study 1 provide important practical guidance for content creators, the correlational nature of our secondary dataset means we cannot fully exclude endogenous influences. Accordingly, we conduct experiments (Studies 2, 3, and 4) to directly manipulate atypicality and test its causal impact on engagement.

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## Study 2: Experimentally Manipulating Atypicality

Study 2 experimentally manipulates atypicality, allowing for a clean test of its causal effect on engagement.

### *Method*

Two hundred one Prolific participants ($M_{age}$ = 38.43, $SD$ = 12.23; 58.71% female) completed this study and were randomly assigned to one of two between-subjects conditions: atypical or typical content.

All participants viewed six TikTok dance videos from different creators of the trend "What You Know 'Bout Love." Five were publicly available videos from TikTok, in which the creators all performed the same dance choreography (i.e., followed the trend closely). The

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remaining video (positioned as Video 4) was the focal video, produced for this research by a professional content creator.

The only difference between conditions was the atypicality of the focal video: In the typical condition, the choreography was the same as in the other videos. In the atypical condition, however, the creator performed a different choreography (screenshots in Web Appendix E). Everything else (e.g., the creator, music snippet, background, and creator's outfit) was held constant. As expected, a pre-test ($N = 62$) found no difference between conditions regarding dance or video quality (both $p > .3$; details in Web Appendix E). All videos were placed on the same survey page, so participants could go back and forth when watching.

After watching the videos, participants completed the dependent variable (i.e., which video they would choose to like or comment on). Finally, participants filled out a brief post-questionnaire including a manipulation check ("To which degree did ALL videos (without any exception) perform the same dance?"; 1 = not at all (at least one video differed), 7 = very much so (no video differed)), questions regarding their social media usage and technical issues while watching the videos, and demographics. Supporting the effectiveness of the manipulation, participants in the atypical condition were less likely to perceive all videos as performing the same dance ($M_{atypical} = 3.59$, $SD = 2.21$; $M_{typical} = 5.92$, SD = 1.28; $t(199) = 9.17$, $p < .001$). Screenshots of all videos and further information regarding the study can be found in Web Appendix E.

***Results***

Consistent with the field data, participants were more interested in engaging with the focal video when it was atypical (24% vs. 12.87%; $\chi^2(1, N = 201) = 4.14$, $p = .042$, *Cramér's V* =

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.144). This holds controlling for frequency of social media usage ($\beta_{atypical}$ = .750, $p$ = .048).

***Discussion***

Results of Study 2 underscore the findings of the field data. By experimentally manipulating atypicality, it provides direct causal evidence that atypicality increases engagement. Participants were nearly twice as likely to indicate engagement when the video was atypical (24% vs. 12.87%).

A follow-up study (detailed in Web Appendix E) replicated Study 2 using different focal and control videos across three different ranges of atypicality. The results indicate that an atypical video creates more engagement, even when deviations are only moderate.

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**Study 3: Experimentally Testing the Role of Trend Exposure**

To further test atypicality's influence on engagement, Study 3 examines the moderating role of trend exposure. As discussed in Study 1, people first need to be aware of a pattern (i.e., a trend) before they can recognize a version of it is atypical. We manipulated the number of videos from the focal trend that participants see, and examined whether, consistent with the field data, atypicality is more likely to increase engagement when it should be easier to observe (i.e., when viewers are exposed more to the trend).

***Method***

Four hundred ninety-four Prolific participants ($M_{age}$ = 39.97, SD = 12.81; 56.48% female) completed the study and were assigned condition in a 2 (atypical vs. typical video) × 2 (trend

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exposure: high vs. low) between-subjects design.

Study 3 used a similar procedure to Study 2. Atypicality was manipulated using the same pre-tested focal videos, and trend exposure was manipulated through whether participants were exposed to other trend videos first. In the high-exposure condition, as in Study 2, participants saw five videos that followed the trend. In the low exposure condition, however, we used videos from the same content creators that were unrelated to the focal trend (details in Web Appendix E). Consequently, in the low exposure condition, the focal videos' atypicality should be less clear.

As in Study 2, participants indicated their choice of video to engage with as the primary dependent variable. Then, they completed a post-questionnaire, including a manipulation check for trend exposure ("To which degree did the videos all perform the same choreography to the same music?"; 1 = not at all (at least one video differed), 7 = very much so (no video differed)). Confirming the manipulation's effectiveness, videos in the high-exposure condition were seen as more similar to each other than in the low exposure condition ($M_{\text{low exposure}}$ = 2.69, SD = 1.82; $M_{\text{high exposure}}$ = 5.05, SD = 1.72; $t(492) = -14.86$, $p < .001$).

To further validate the manipulation, we used the MUVID app to calculate the within-set atypicality (average pairwise cosine distance) of the five reference videos shown in each condition. The high-exposure reference sets showed significantly lower within-set distance ($M$ = .195, $SD$ = .154) compared to the low-exposure sets ($M$ = .405, $SD$ = .149; $t(18) = -3.10$, $p$ = .006). This confirms that our manipulation successfully created distinct evaluative contexts: videos were more similar to each other in the high-exposure condition than in the low-exposure one.

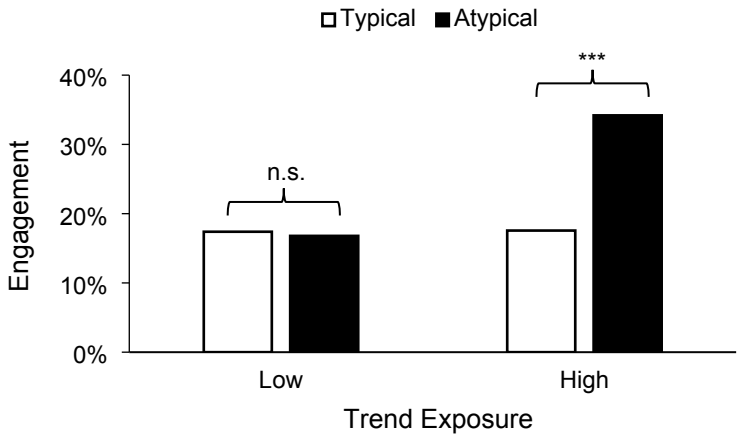
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***Results***

Consistent with the field data, results of a logistic regression reveal an atypicality × trend exposure interaction ($\beta = .934$, $p = .040$). Separate chi-square tests confirm these results: When trend exposure was high, atypicality increased engagement with the focal video (34.19% vs. 17.56%; $\chi^2(1, N = 248) = 9.02$, $p = .003$, *Cramér's V* = .191). When trend exposure was low, however, this effect disappeared (16.79% vs. 17.39%; $\chi^2(1, N = 246) = .015$, $p = .901$, *Cramér's V* = .008; Figure 4).

Figure 4: Moderation by Trend Exposure




*Notes:* Bars represent the proportion of participants (in %) selecting the focal video, as the operationalization of engagement.

***Discussion***

Results of Study 3 underscore atypicality's causal impact and provide further support for the interaction found in Study 1. As in the field data, atypical videos were more likely to increase engagement when atypicality was easier to observe (i.e., people were more exposed to the trend).

A follow-up experiment (see Web Appendix E) replicated this finding and tested potential explanations using open- and closed-ended questions. Rather than typical videos becoming boring with increased exposure, results suggest that atypicality positively affects

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engagement primarily because atypical videos are perceived as more unique, and their creators as

## Study 4: Field Experiment on TikTok

While results of Studies 2 and 3 provide strong causal evidence that atypicality boosts engagement, one could still wonder whether the effect holds in a more naturalistic setting. To test this possibility, we conducted a field experiment. We randomly assigned real TikTok creators to produce both typical and atypical interpretations of trending dances, which they posted to the corresponding trends on TikTok. We then measured the resulting impact on actual engagement.

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### *Method*

We recruited content creators via creator platforms to produce videos that aligned with current TikTok trends and to post them on their personal accounts. The 38 creators in our sample were active TikTok users, with an average follower count of 41,754 (min = 10, max = 949,280). We selected five major trends based on the TikTok Creator Center (see Web Appendix F for an overview), and creators were randomly assigned to two out of the five. We used this design to avoid the possibility that posting multiple videos within the same trend could reduce engagement or that the effect could hold only within one specific trend.

Each creator was instructed to produce a video for each of the two trends assigned to them—one in a typical style and one in an atypical one. We randomized *which* of the two trends they should interpret typically and which atypically, and creators were instructed to keep everything else (e.g., background, camera angle, outfit, and dance quality) as constant as

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possible.

To objectively validate that the creators successfully produced "typical" and "atypical" content as instructed, we extracted the visual embeddings of all creator-produced videos using the MUVID app. We then computed an atypicality score for each video by calculating its average cosine distance to all other existing videos within our sample assigned to the same trend. A t-test confirms that videos in the atypical condition had higher atypicality scores ($M$ = .316, $SD$ = .08) than those in the typical condition ($M$ = .280, $SD$ = .07; $t$ = -2.42, $p$ = .018).

Note that the manipulation did not inadvertently alter production value. There were no significant differences between the atypical and typical conditions regarding visual quality ($M_{atypical}$ = 4.76, $M_{typical}$ = 4.73; $t$ = -.86, $p$ = .39) or dance quality ($M_{atypical}$ = 4.63, $M_{typical}$ = 4.58; $t$ = -.64, $p$ = .53). These checks confirm that our manipulation successfully isolated atypicality without introducing quality-related confounds.

Finally, we tracked engagement. To mitigate potential algorithmic bias stemming from simultaneous posting, creators were instructed to post the two videos approximately three days apart. To control for potential timing effects, the posting order (atypical first vs. typical first) was counterbalanced between creators. On average, the posted videos received 783 views (min = 20, max = 5,375) and 59 likes (min = 1, max = 333).

We estimate a similar grouped logit model as in Study 1 (Agresti 2013). Because each creator contributes two observations, the error terms may be correlated within creators. We therefore cluster standard errors at the creator level and include creator-fixed effects, such that the treatment effect is identified from within-creator comparisons.

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***Results***

Results indicate that atypicality boosts engagement ($\beta = .093$; $SE = .046$; $t = 2.01$; $p = .044$), with atypical videos generating more engagement than typical ones. Interpreted as an odds ratio, the estimate corresponds to approximately a 10% increase in the odds that a view results in a like.

***Discussion***

The field experiment corroborates the results of the prior studies and highlights their practical utility. Even in a naturalistic setting, with actual content creators, more atypical videos receive higher engagement. These findings suggest that introducing atypical elements into dance trend videos can successfully increase viewer engagement in the real world.

Note that our design (i.e., randomized treatments, pairing two videos within the same creator, and analyzing engagement conditional on exposure) addresses audience and platform effects that are often confounding factors in ad A/B testing on social media platforms (Braun and Schwartz 2025). Further, the convergence of results across secondary data, online experiments, and a field experiment provides strong evidence for both the internal and external validity of the observed relationship.

Next we discuss the MUVID app.

## App Implementation: The MUVID Web App

Beyond serving merely as a data extraction tool, MUVID acted as the methodological backbone of our empirical strategy, enabling us to quantify crucial video characteristics at scale.

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Specifically, we used MUVID for the following critical steps:

First, to ensure the analysis focused on the correct type of videos, we utilized MUVID's transfer learning capabilities to train a binary classifier that automatically distinguished dance from non-dance videos. Second, TikTok trends are defined by specific audio tracks, and by extracting each video's audio embeddings, the model allowed us to precisely identify and cluster videos into distinct trends. Third, and most critically, to objectively measure atypicality, MUVID's visual embeddings allowed us to calculate the high-dimensional cosine distance between a focal video's visual embedding and those of the trend's prior videos (averaged across videos). Fourth, to effectively control for confounding factors that human coding could not scale to capture, we used the model to generate transfer learning predictions for control variables such as "dance quality" and "visual quality."

We also used MUVID to objectively validate our experimental manipulations. In Study 4, the app's atypicality metric confirmed that the atypical stimuli were indeed significantly distinct from typical ones and its quality scores ruled out potential confounds, demonstrating the app's utility as a manipulation check for behavioral research.

***Features of the MUVID Research App***

To make MUVID's capabilities accessible to researchers and practitioners without deep technical expertise, we designed the MUVID research app, a web-based converter tool. While comparable tools exist for text (e.g., TextAnalyzer.org, Berger, Sherman, and Ungar 2020; WritingClarityCalculator.com, Warren et al. 2021), fewer accessible tools are currently available for extracting embeddings from short-form social media videos. Providing this app should help to stimulate social media video research, as the app has an intuitive click-through web interface

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that allows users to easily extract visual and audio embeddings. Once videos are uploaded, the raw embeddings can be exported as CSV files, enabling flexible further analysis.

Beyond extracting raw data, the app features built-in analytical tools that allow users to directly work with these embeddings. Similar to our own research approach, users can measure similarity based on visual and audio embeddings natively within the interface. First, they can visualize videos by applying principal component analysis (PCA) to the embeddings and projecting them into a two-dimensional space. The app displays pairwise similarities, grouping more similar videos closer to each other. Crucially, the app calculates the atypicality score for each video relative to the specific batch of uploaded videos, such that researchers can define the context themselves. For instance, when uploading a set of videos from a single trend, the resulting scores will reflect within-trend atypicality. However, it is important to note that the raw visual embeddings capture the holistic visual scene, including background, lighting, and colors. Consequently, videos may occasionally be grouped by setting (e.g., a similar background wall) rather than movement content. Second, for a focal video, the app shows the closest match based on cosine distances from a built-in sample of 100,000 videos. Users can also weigh similarity predictions based on audio and visual embeddings.

Other use cases involve predictive modeling using transfer learning based on the video embeddings. This approach has been successfully used in marketing research for analyzing image and text data, such as examining how products or brands in images can affect engagement (e.g., Hartmann et al. 2021; Liu, Dzyabura, and Mizik 2020). Such techniques can also be applied to MUVID's video-based embeddings. To demonstrate this, we implemented several exemplary prediction models that are based on transfer learning in the app. The first model allows users to predict the level of engagement an uploaded video would receive on social

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media. Aside from academic use cases, such a model could help creators to make informed decisions about what to post. Two additional models estimate the level of Visual and Dance Quality (similar to what we used in our Study 1). The MUVID app also includes a hashtag recommender, which suggests hashtags that are likely to be associated with the content and resonate with the target audience. Finally, we integrated a category classifier, which classifies videos into 14 different categories. This function could help marketing researchers filter datasets for specific categories.

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***Potential Applications in Marketing Research***

Aside from our pre-built function, extracted embeddings can help address a range of research questions. Future work could extend MUVID to predict influencer authenticity, for example, based on their appearance in videos. Alternatively, it could use MUVID embeddings to examine how video characteristics shape brand image, detect how product placements influence visibility, or focus on textual elements to examine the interaction between text and visual video elements (as was recently done for text and images; Farace et al. 2025). Table 3 presents possible research questions that could be addressed using extracted embeddings from the MUVID app.

Notably, to derive interpretable features (such as authenticity, or to classify products in videos) other than those provided in the app, one can use transfer learning to train a model based on the extracted embeddings. Specifically, researchers can treat the MUVID embeddings as high-dimensional input features for a lightweight classifier (e.g., logistic regression, or a tree-based model). By training such a classifier on a smaller, labeled dataset of their own (e.g., 1,000 labeled videos), researchers can efficiently fine-tune the model to detect novel constructs, such as “humor,” “creativity,” or “authenticity” without the need for massive computational resources

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Table 3: Potential Research Questions That Can Be Tackled with the MUVID App

| **Research Questions Beyond Current Investigation** |
|---|
| 1) Which video characteristics influence the likelihood of a dance to become a viral trend? |
| 2) How can a composition of audio and visual components convey a certain brand image (e.g., certain music, rapid scene changes, fast versus smooth movements)? |
| 3) How do body movement and body language influence the perceived authenticity of an influencer? |
| 4) What is the optimal temporal and spatial placement of brand logos or products within videos to maximize visibility, memorability, and overall campaign effectiveness? |
| 5) How do temporal dynamics (e.g., scene changes) affect viewer attention, interest, and overall engagement? |
| 6) Which sorts of context-specific features (lenses, social media filters) predict a video's engagement? |
| 7) How can video embeddings be made more explainable? |
| 8) Which video components make a trend go viral from an observer perspective (e.g., arousing movements, video-audio combinations, etc.)? Are there differences between platforms (TikTok, YouTube Shorts, Instagram Reels) and trend styles? |
| 9) Which video components motivate users to adopt a trend (e.g., mental simulation processes, the degree of difficulty of movements)? |
| 10) Which video characteristics are most predictive of the success of an influencer channel? |

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## General Discussion

The growing importance of video content in social media, especially short videos, has led brands and content creators to update their content creation strategies. Trend following is a widely employed strategy to generate engagement, particularly in the fast-paced world of social media, where trends can emerge and fade rapidly (Israeli et al. 2023). Nevertheless, trend following has, until now, received limited research attention. By developing and applying MUVID, a multimodal unsupervised video analyzer, we take a first step toward addressing this gap.

Analysis of tens of thousands of TikTok videos using our MUVID app reveals that videos that deviate from the trend tend to receive higher engagement, and experiments underscore atypicality's causal impact. Results also indicate that the relationship becomes stronger as more trend videos accumulate (i.e., as audience exposure to the trend increases), and deviations in the final part of the video are more strongly associated with engagement.

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***Theoretical Contributions***

Our research contributes to current social media research both conceptually and methodologically. Conceptually, our work extends knowledge about video consumption and social media engagement by examining the context of dance trend following. Prior research suggests that users often prefer to watch content from the same category (Woolley and Sharif 2022) and that easier-to-process videos receive higher engagement (Stuppy, Landwehr, and McGraw 2024). With dance trends, however, variation receives more engagement, likely because atypicality generates interest (Berger and Packard 2018). These results suggest that the effects of novelty and familiarity depend on the context and degree of prior exposure to content patterns.

Methodologically, our work contributes to the literature on unstructured data analyses in social media (Dang et al. 2026; Hartmann et al. 2019; Liu, Dzyabura, and Mizik 2020). With MUVID, we introduce a multimodal and unsupervised learning approach to extract temporal and dynamic aspects of short social media videos (e.g., Ramanathan et al. 2015). The resulting embeddings are versatile for research questions involving video analytics, as they are a starting point for transfer learning or similarity measurement (Gabel, Guhl, and Klapper 2019; Timoshenko and Hauser 2019).

***Managerial Implications***

Our work has a variety of practical implications. Brands and content creators hoping to engage audiences through trending content, for example, should consider producing more atypical trending dance videos, especially as users become more exposed to a trend. Evidence

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also suggests that deviating later in a video, rather than at the beginning, may be associated with stronger engagement — a pattern that warrants further experimental investigation.

Similarly, platforms can use our findings to tailor content recommendations according to the phases of trending content. Algorithms may promote more similar videos for emerging trends, but then suggest more deviating videos as the trend matures. Platforms could also use our insights to help content creators understand what to create (e.g., the TikTok Creative Center gives concrete guidelines to content creators). From a broader perspective, our research suggests considering the similarity of one piece of content to whatever is out there already, which can be done using applications like our research app.

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***Limitations and Future Research***

As with any new area, there are opportunities for future research. While we examine existing trends, for example, future research could investigate what makes a trend go viral in the first place (e.g., using video analytics to distinguish between characteristics of trending and non-trending videos). Future work could also study specific forms of atypicality (e.g., group dances, personalities, special effects) and which mechanisms drive atypicality's effect (e.g., attention, surprise, or social inferences regarding the creators; Elsbach and Kramer 2003). One could also examine whether atypicality has positive effects for other types of trends, such as transformation challenges with makeovers, or brand-specific challenges, like the Vogue challenge, where users could put themselves on Vogue magazine's cover (Okwodu 2020).

Methodologically, while MUVID offers a powerful and scalable approach to analyzing unstructured video content, the model has the typical limitations of deep learning approaches. Specifically, explainability remains a challenge. Future research could address this gap by

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integrating attention mechanisms that highlight the most influential parts of a video (Dosovitskiy et al. 2020; Wals et al. 2026). This would not only improve the utility of the tool for content creators but also contribute to the broader goal of making AI-driven analytics more transparent. Importantly, when applying MUVID to new contexts via transfer learning procedures, we strongly recommend evaluating the model's performance on relevant downstream tasks (Schiappa, Rawat, and Shah 2023). Further, researchers should be aware that MUVID was trained specifically on social media short videos, which means that it will likely perform best for the categories that were most prevalent in our training sample (e.g., Ceci 2022; see Web Appendix B). Finally, if highly specific features are the object of interest—if, say, the aim is to gain a fine-grained understanding of background atypicality or nuanced temporal evolutions in audio tracks—a more specialized, domain-specific training process may be necessary. These boundaries apply equally to the interactive web app, which relies on MUVID as its underlying model.

Looking ahead, we envision MUVID as a modular and extensible framework rather than a fixed solution: future versions can incorporate newer video encoders and multimodal foundation models as the landscape evolves, ensuring the tool remains relevant as the field advances. More broadly, agentic AI represents a promising direction for making video embedding research more accessible (Yoo et al. 2025). Intelligent agents could serve as methodological guides — helping researchers select appropriate models, preprocess videos, generate task-specific embeddings, evaluate representation quality, and recommend validation checks — thereby enabling more transparent and responsible application of video analytics across diverse research contexts (similar to LLM usage for research proposed by Arora et al. 2025 or Blanchard et al. 2025).

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***Conclusion***

Trend following is an important phenomenon, and video analytics is a powerful tool to investigate what drives success. This research not only advances our understanding of what makes trend-following content successful, but also provides the tools needed to apply video analytics in research and practice. We hope that MUVID will catalyze further research and development in the field, paving the way for innovations in the analysis of video content.

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Author Accepted Manuscript

## References


Aggarwal, Charu C. (2017), "An Introduction to Outlier Analysis," in *Outlier Analysis*, Cham: Springer International Publishing, 1–34.

Agresti, Alan (2013), *Categorical Data Analysis*, John Wiley & Sons.

Al-Tahan, Haider and Yalda Mohsenzadeh (2021), "CLAR: Contrastive Learning of Auditory Representations," in *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics*, Proceedings of Machine Learning Research, Arindam Banerjee and Kenji Fukumizu, eds., PMLR, 2530–38.

Angiulli, Fabrizio and Clara Pizzuti (2002), "Fast outlier detection in high dimensional spaces," in *Lecture Notes in Computer Science*, Lecture notes in computer science, Berlin, Heidelberg: Springer Berlin Heidelberg, 15–27.

Arora, Neeraj, Ishita Chakraborty, and Yohei Nishimura (2025), "AI–human hybrids for marketing research: Leveraging large language models (LLMs) as collaborators," *Journal of Marketing*, 89 (2), 43–70.

Aspire (2024), "The State of Influencer Marketing 2024," *Aspire*, (accessed February 17, 2026), https://learn.aspireiq.com/rs/982-DON-266/images/The_State_of_Influencer_Marketing_2024.pdf?version=3.

Beichert, Maximilian, Andreas Bayerl, Jacob Goldenberg, and Andreas Lanz (2024), "Revenue Generation Through Influencer Marketing," *Journal of Marketing*, 88 (4), 40–63.

Berger, Jonah (2011), "Arousal increases social transmission of information," *Psychological Science*, 22 (7), 891–93.

Berger, Jonah (2014), "Word of mouth and interpersonal communication: A review and directions for future research," *Journal of Consumer Psychology,* 24 (4), 586–607.

Berger, Jonah, Ashlee Humphreys, Stephan Ludwig, Wendy W. Moe, Oded Netzer, and David A. Schweidel (2020), "Uniting the Tribes: Using Text for Marketing Insight," *Journal of Marketing*, 84 (1), 1–25.

Berger, Jonah and Raghuram Iyengar (2013), "Communication channels and word of mouth: How the medium shapes the message," *Journal of Consumer Research*, 40 (3), 567–79.

Berger, Jonah and Katherine L. Milkman (2012), "What makes online content viral?," *Journal of Marketing Research,* 49 (2), 192–205.

Berger, Jonah and Grant Packard (2018), "Are Atypical Things More Popular?," *Psychological Science*, 29 (7), 1178–84.

Berger, Jonah and Eric M. Schwartz (2011), "What Drives Immediate and Ongoing Word of Mouth?," *Journal of Marketing Research*, 48 (5), 869–80.

Berger, Jonah, Garrick Sherman, and Lyle Ungar (2020), "TextAnalyzer," https://textanalyzer.org.

Blanchard, Simon J., Nofar Duani, Aaron M. Garvey, Oded Netzer, and Travis Tae Oh (2025), "New Tools, New Rules: A Practical Guide to Effective and Responsible Generative AI Use for Surveys and Experiments in Research," *Journal of Marketing*, 89 (6), 119–39.

Borah, Abhishek, Sourindra Banerjee, Yu-Ting Lin, Apurv Jain, and Andreas B. Eisingerich (2020), "Improvised Marketing Interventions in Social Media," *Journal of Marketing*, 84 (2), 69–91.

Braun, Michael and Eric Schwartz (2025), "Where A/B Testing Goes Wrong: How Divergent Delivery Affects What Online Experiments Cannot (and Can) Tell You About How Customers Respond to Advertising," *Journal of Marketing,* 89 (2), 71–95.


For Review Only

Author Accepted Manuscript

Burgoon, Judee K. (1993), "Interpersonal Expectations, Expectancy Violations, and Emotional Communication," *Journal of Language and Social Psychology*, 12 (1–2), 30–48.

Büschken, Joachim, Thomas Otter, and Greg M. Allenby (2025), "Accounting for Formative and Reflective Topics in Product Review Data for Better Consumer Insights," *Journal of Marketing Research*, 62 (6), 1045–62.

Cardillo, Anthony (2026), "Top 100 Searches On TikTok (2025)," *Exploding Topics*, (accessed March 17, 2026), https://explodingtopics.com/blog/popular-searches-on-tiktok.

Cascio Rizzo, Giovanni Luca, Jonah Berger, Matteo De Angelis, and Rumen Pozharliev (2023), "How Sensory Language Shapes Influencer's Impact," *Journal of Consumer Research*, 50 (4), 810–25.

Cascio Rizzo, Giovanni Luca, Francisco Villarroel Ordenes, Rumen Pozharliev, Matteo De Angelis, and Michele Costabile (2024), "How high-arousal language shapes micro-versus macro-influencers' impact." *Journal of Marketing,* 88 (4), 107–28.

Ceci, Laura (2022), "Most popular content categories on TikTok worldwide as of July 2020, by number of hashtag views," *Statista*, (accessed April 16, 2023), https://www.statista.com/statistics/1130988/most-popular-categories-tiktok-worldwide-hashtag-views/.

Chakraborty, Ishita, Khai Chiong, Howard Dover, and K. Sudhir (2025), "Can AI and AI-hybrids detect persuasion skills? Salesforce hiring with conversational video interviews," *Marketing Science*, 44 (1), 30–53.

Cheng, Yan, Jingbo Wang, Xinyu Cao, Zuo-Jun Max Shen, and Yuhui Zhang (2026), "A Deep-DiD Method to Estimate Heterogeneous Treatment Effects: Application to Content Creator Selection," *Marketing Science*, forthcoming.

Colicev, Anatoli, Ashwin Malshe, Koen Pauwels, and Peter O'Connor (2018), "Improving Consumer Mindset Metrics and Shareholder Value through Social Media: The Different Roles of Owned and Earned Media," *Journal of Marketing*, 82 (1), 37–56.

Dang, Ivy Chu, Canice M. C. Kwan, Jayson S. Jia, and Yang Shi (2026), "When Words Meet Visuals: How Content Composition Drives Social Media Engagement for Marketer-Generated Content," *Journal of Marketing Research*, 63 (1), 167–90.

Dellatto, Marisa (2021), "TikTok Hits 1 Billion monthly active users," *Forbes*, (accessed December 16, 2022), https://www.forbes.com/sites/marisadellatto/2021/09/27/tiktok-hits-1-billion-monthly-active-users.

Dencheva, Valentina (2023), "Types of most engaging content for social media marketing according to consumers in the United States as of April 2022," *Statista*, (accessed July 6, 2023), https://www.statista.com/statistics/1273876/content-social-media-marketing-usa/.

Dixon, Stacy J. (2022), "Social media use during COVID-19 worldwide - Statistics & Facts," *Statista*, (accessed June 30, 2023), https://www.statista.com/topics/7863/social-media-use-during-coronavirus-covid-19-worldwide.

Dosovitskiy, Alexey, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby (2020), "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," *arXiv [cs.CV]*.

Elsbach, Kimberly D. and Roderick M. Kramer (2003), "Assessing creativity in Hollywood pitch meetings: Evidence for a dual-process model of creativity judgments," *Academy of Management Journal*, 46 (3), 283–301.

Elsen, Millie, Rik Pieters, and Michel Wedel (2016), "Thin slice impressions: How advertising

Peer Review Version

Author Accepted Manuscript

evaluation depends on exposure duration," *Journal of Marketing Research,* 53 (4), 563–79.

Ester, Martin, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu (1996), "A density-based algorithm for discovering clusters in large spatial databases with noise," aaai.org, https://www.aaai.org/Papers/KDD/1996/KDD96-037.pdf.

Farace, Stefania, Francisco Villarroel Ordenes, Dennis Herhausen, Dhruv Grewal, and Ko de Ruyter (2025), "Standing out while fitting in: Visual design of text overlays in social media communication," Journal of Marketing. https://doi.org/10.1177/00222429251322773.

Feng, Xiaohang, Shunyuan Zhang, Xiao Liu, Kannan Srinivasan, and Cait Lamberton (2025), "An AI method to score celebrity visual potential," *Journal of Marketing Research*, 62 (5), 757–75.

Follet, Gillian (2023), "5 New TikTok Trends That Brands Can Follow," *AdAge*, (accessed April 17, 2023), https://adage.com/article/digital-marketing-ad-tech-news/5-new-tiktok-trends-brands-can-follow/2468326.

Gabel, Sebastian, Daniel Guhl, and Daniel Klapper (2019), "P2V-MAP: Mapping Market Structures for Large Retail Assortments," *Journal of Marketing Research,* 56 (4), 557–80.

Graffius, Scott (2025), "Lifespan (Half-Life) of Social Media Posts: Update for 2025," *Scott Graffius*, (accessed July 23, 2025), https://www.scottgraffius.com/blog/files/lifespan-halflife-of-social-media-posts-update-for-2025.html.

Grewal, Rajdeep, Sachin Gupta, and Rebecca Hamilton (2021), "Marketing Insights from Multimedia Data: Text, Image, Audio, and Video," *Journal of Marketing Research,* 58 (6), 1025–33.

Hartmann, Jochen, Juliana Huppertz, Christina Schamp, and Mark Heitmann (2019), "Comparing automated text classification methods," *International Journal of Research in Marketing*, 36 (1), 20–38.

Hartmann, Jochen, Mark Heitmann, Christina Schamp, and Oded Netzer (2021), "The Power of Brand Selfies," *Journal of Marketing Research,* 58 (6), 1159–77.

He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun (2016), "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, openaccess.thecvf.com, 770–78.

Hershey, Shawn, Sourish Chaudhuri, Daniel P. W. Ellis, Jort F. Gemmeke, Aren Jansen, R. Channing Moore, Manoj Plakal, Devin Platt, Rif A. Saurous, Bryan Seybold, Malcolm Slaney, Ron J. Weiss, and Kevin Wilson (2017), "CNN architectures for large-scale audio classification," in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 131–35.

Huang, Justin, Rupali Kaul, and Sridhar Narayanan (2025), "Novelty in Content Creation: Experimental Results Using Deep Learning–Based Image Recognition on a Large Social Network," *Marketing Science*, 45 (2), 335–58.

Influencer Marketing Hub (2022), "The Best Of Brands Using TikTok In 2022: What Are Their Secrets?," *Influencer Marketing Hub*, (accessed January 6, 2023), https://influencermarketinghub.com/best-brands-using-tiktok/.

Israeli, Ayelet, Leonard Schlesinger, and Matt Higgins (2023), "How to Seed Organic Marketing in a Video-First World," (accessed June 24, 2024), https://hbr.org/2023/02/how-to-seed-organic-marketing-in-a-video-first-world.

Jansen, Nora, Oliver Hinz, Clemens Deusser, and Thorsten Strufe (2021), "Is the Buzz on? – A Buzz Detection System for Viral Posts in Social Media," *Journal of Interactive Marketing*, 56 (1), 1–17.

For Review Purposes

Author Accepted Manuscript

Kanuri, Vamsi K., Yixing Chen, and Shrihari Sridhar (2018), "Scheduling Content on Social Media: Theory, Evidence, and Application," *Journal of Marketing*, 82 (6), 89–108.

Klug, Daniel, Yiluo Qin, Morgan Evans, and Geoff Kaufman (2021), "Trick and Please. A Mixed-Method Study On User Assumptions About the TikTok Algorithm," in *Proceedings of the 13th ACM Web Science Conference 2021*, WebSci '21, New York, NY, USA: Association for Computing Machinery, 84–92.

Kübler, Raoul V., Anatoli Colicev, and Koen H. Pauwels (2020), "Social Media's Impact on the Consumer Mindset: When to Use Which Sentiment Extraction Tool?," *Journal of Interactive Marketing*, 50 (1), 136–55.

Labroo, Aparna A. and Anastasiya Pocheptsova (2016), "Metacognition and consumer judgment: fluency is pleasant but disfluency ignites interest," *Current Opinion in Psychology*, 10, 154–59.

Lamberton, Cait and Andrew T. Stephen (2016), "A thematic exploration of digital, social media, and mobile marketing: Research evolution from 2000 to 2015 and an agenda for future inquiry," *Journal of Marketing*, 80 (6), 146–72.

Lambrecht, Anja, Catherine Tucker, and Caroline Wiertz (2018), "Advertising to Early Trend Propagators: Evidence from Twitter," *Marketing Science*, 37 (2), 177–99.

Landwehr, Jan R., Aparna A. Labroo, and Andreas Herrmann (2011), "Gut Liking for the Ordinary: Incorporating Design Fluency Improves Automobile Sales Forecasts," *Marketing Science*, 30 (3), 416–29.

Lang, Kirsti (2025), "How Often Should You Post on TikTok in 2026? Data From 11 Million+ Posts," *Buffer*, (accessed March 18, 2026), https://buffer.com/resources/how-often-should-you-post-on-tiktok/.

Li, Yiyi and Ying Xie (2020), "Is a Picture Worth a Thousand Words? An Empirical Study of Image Content and Social Media Engagement," *Journal of Marketing Research,* 57 (1), 1–19.

Lin, Yan, Dai Yao, and Xingyu Chen (2021), "Happiness Begets Money: Emotion and Engagement in Live Streaming," *Journal of Marketing Research,* 58 (3), 417–38.

Liu, Liu, Daria Dzyabura, and Natalie Mizik (2020), "Visual listening in: Extracting brand image portrayed on social media," *Marketing Science*, 39 (4), 669–86.

Loewenstein, Jeffrey, Rajagopal Raghunathan, and Chip Heath (2011), "The Repetition-Break Plot Structure Makes Effective Television Advertisements," *Journal of Marketing*, 75 (5), 105–19.

Lu, Shasha, Li Xiao, and Min Ding (2016), "A Video-Based Automated Recommender (VAR) System for Garments," *Marketing Science*, 35 (3), 484–510.

Lysyakov, Mikhail, P. K. Kannan, Siva Viswanathan, and Kunpeng Zhang (2025), "Retailer Differentiation in Social Media: An Investigation of Firm-Generated Content on Twitter," *Journal of Marketing*, 89 (4), 39–58.

Meire, Matthijs, Kelly Hewett, Michel Ballings, V. Kumar, and Dirk Van den Poel (2019), "The Role of Marketer-Generated Content in Customer Engagement Marketing," *Journal of Marketing*, 83 (6), 21–42.

Metricool (2023), "Average TikTok video length from March 2023 to August 2023, by number of video views," *Statista*, (accessed June 4, 2026), https://www.statista.com/statistics/1372569/tiktok-video-duration-by-number-of-views/.

Moldovan, Sarit, Jacob Goldenberg, and Amitava Chattopadhyay (2011), "The different roles of product originality and usefulness in generating word-of-mouth," *International Journal of*

For Review Only

Author Accepted Manuscript

*Research in Marketing*, 28 (2), 109–19.

Monahan, Jennifer L., Sheila T. Murphy, and Robert B. Zajonc (2000), "Subliminal mere exposure: Specific, general, and diffuse effects," *Psychological Science*, 11 (6), 462–66.

Okwodu, Janelle (2020), "The #VogueChallenge Is More Than a Hashtag," *Vogue*, (accessed July 21, 2023), https://www.vogue.com/article/vogue-cover-challenge-more-than-hashtag.

Olsson, Sam (2023), "40 TikTok Key Statistics and Trends for 2024," *Kurve Blog*, (accessed February 20, 2026), https://kurve.co.uk/blog/tiktok-ad-statistics.

Overgoor, Gijs, William Rand, Willemijn van Dolen, and Masoud Mazloom (2022), "Simplicity is not key: Understanding firm-generated social media images and consumer liking," *International Journal of Research in Marketing*, 39 (3), 639–55.

Packard, Grant and Jonah Berger (2024), "The Emergence and Evolution of Consumer Language Research," *Journal of Consumer Research*, 51 (1), 42–51.

Qian, Rui, Tianjian Meng, Boqing Gong, Ming-Hsuan Yang, Huisheng Wang, Serge Belongie, and Yin Cui (2021), "Spatiotemporal contrastive video representation learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, openaccess.thecvf.com, 6964–74.

Rajaram, Prashant and Puneet Manchanda (2023), "Video Influencers: Unboxing the Mystique," *Available at SSRN 3752107*.

Ramanathan, Vignesh, K. Tang, Greg Mori, and Li Fei-Fei (2015), "Learning temporal embeddings for complex video analysis," *IEEE International Conference on Computer Vision*, 4471–79.

Satenstein, Liana (2021), "This Wildly Popular Dance of the '00s Is Back With Bad Bunny," *Vogue*, (accessed October 29, 2024), https://www.vogue.com/article/vogue-recreates-the-macarena-with-bad-bunny.

Schiappa, Madeline C., Yogesh S. Rawat, and Mubarak Shah (2023), "Self-Supervised Learning for Videos: A Survey," *ACM Computing Surveys*, 55 (13s), 1–37.

Schweidel, David A. and Wendy W. Moe (2016), "Binge watching and advertising," *Journal of Marketing*, 80 (5), 1–19.

Schwenzow, Jasper, Jochen Hartmann, Amos Schikowsky, and Mark Heitmann (2021), "Understanding videos at scale: How to extract insights for business research," *Journal of Business Research*, 123, 367–79.

Shin, Donghyuk, Shu He, Gene Moo Lee, Andrew B. Whinston, Suleyman Cetintas, and Kuang-Chih Lee (2020), "Enhancing social media analysis with visual data analytics: A deep learning approach," *MIS Quarterly*, 44 (4), 1459–92.

Shriver, Scott K., Harikesh S. Nair, and Reto Hofstetter (2013), "Social Ties and User-Generated Content: Evidence from an Online Social Network," *Management Science*, 59 (6), 1425–43.

Stuppy, Anika, Jan R. Landwehr, and A. Peter McGraw (2024), "The art of slowness: Slow motion enhances consumer evaluations by increasing processing fluency," *Journal of Marketing Research,* 61 (2), 185–203.

Teixeira, Thales, Rosalind Picard, and Rana el Kaliouby (2014), "Why, When, and How Much to Entertain Consumers in Advertisements? A Web-Based Facial Tracking Field Study," *Marketing Science*, 33 (6), 809–27.

Teixeira, Thales S., Michel Wedel, and Rik Pieters (2010), "Moment-to-Moment Optimal Branding in TV Commercials: Preventing Avoidance by Pulsing," *Marketing Science*, 29 (5), 783–804.

For Review Only

Author Accepted Manuscript

Tian, Zijun, Ryan Dew, and Raghuram Iyengar (2024), "Mega or Micro? Influencer Selection Using Follower Elasticity," *Journal of Marketing Research,* 61 (3), 472–95.

Timoshenko, Artem and John R. Hauser (2019), "Identifying Customer Needs from User-Generated Content," *Marketing Science*, 38 (1), 1–20.

Tran, Du, Heng Wang, Lorenzo Torresani, Jamie Ray, Yann LeCun, and Manohar Paluri (2018), "A closer look at spatiotemporal convolutions for action recognition," in *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*, 6450–59.

Villarroel Ordenes, Francisco, Dhruv Grewal, Stephan Ludwig, Ko de Ruyter, Dominik Mahr, and Martin Wetzels (2018), "Cutting through Content Clutter: How Speech and Image Acts Drive Consumer Sharing of Social Media Brand Messages," *Journal of Consumer Research*, 45 (5), 988–1012.

Virós-Martín, Clara, Mireia Montaña-Blasco, and Mònika Jiménez-Morales (2024), "Can't Stop Scrolling! Adolescents' Patterns of TikTok Use and Digital Well-Being Self-Perception," *Humanities and Social Sciences Communications*, 11 (1), 1–11.

Wals, Sammy, Dhruv Grewal, Ines Wilms, Tim Hilken, Alexia Briassouli, and Martin Wetzels (2026), "The dynamic effects of visual complexity and scene cuts on viewer attention," *Journal of the Academy of Marketing Science*, 1–22.

Wang, Tongzhou and Phillip Isola (2020), "Understanding contrastive representation learning through alignment and uniformity on the hypersphere," *International Conference on Machine Learning*, (Hal Daumé III and Aarti Singh, eds.), 9929–39.

Wang, Xin, Neil Bendle, and Yinjie Pan (2024), "Beyond text: Marketing strategy in a world turned upside down," *Journal of the Academy of Marketing Science*, 52 (4), 939–54.

Wang, Yifei, Jui Ramaprasad, and Anandasivam Gopal (2025), "Dancing to the #challenge: The effect of TikTok on closing the artist gender gap in the music industry," *MIS Quarterly*, 49 (3), 861–88.

Warren, Nooshin L., Matthew Farmer, Tianyu Gu, and Caleb Warren (2021), "Marketing Ideas: How to Write Research Articles that Readers Understand and Cite," *Journal of Marketing*, 85 (5), 42–57.

Wathieu, Luc (2004), "Consumer Habituation," *Management Science*, 50 (5), 587–96.

Wei, Yanhao, Jihoon Hong, and Gerard J. Tellis (2022), "Machine Learning for Creativity: Using Similarity Networks to Design Better Crowdfunding Projects," *Journal of Marketing*, 86 (2), 87–104.

Wies, Simone, Alexander Bleier, and Alexander Edeling (2023), "Finding Goldilocks Influencers: How Follower Count Drives Social Media Engagement," *Journal of Marketing*, 87 (3), 383–405.

Woolley, Kaitlin and Marissa A. Sharif (2022), "Down a Rabbit Hole: How Prior Media Consumption Shapes Subsequent Media Consumption," *Journal of Marketing Research,* 59 (3), 453–71.

Yang, Jeremy, Juanjuan Zhang, and Yuhan Zhang (2025), "Engagement that sells: Influencer video advertising on TikTok," *Marketing Science*, 44 (2), 247–67.

Yang, Yi, Kunpeng Zhang, and P. K. Kannan (2022), "Identifying Market Structure: A Deep Network Representation Learning of Social Engagement," *Journal of Marketing*, 86 (4), 37–56.

Yoo, Kiwoong, Michael Haenlein, and Kelly Hewett (2025), "A whole new world, a new fantastic point of view: Charting unexplored territories in consumer research with generative artificial intelligence," *Journal of the Academy of Marketing Science*, 53 (3),

Peer Review Version

Author Accepted Manuscript

723–59.

Zabala, Lucia (2025), "Most popular TikTok Categories: A Deep Dive Into Trends," *Creator Hero*, (accessed February 20, 2026), https://www.creator-hero.com/blog/most-popular-tiktok-categories.

Zhong, Ning and David A. Schweidel (2020), "Capturing Changes in Social Media Content: A Multiple Latent Changepoint Topic Model," *Marketing Science*, 39 (4), 827–46.

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# Web Appendix: How Closely Should You Follow a Trend? Atypicality and Engagement on Social Media

**Marc Bravin**
Lucerne University of Applied Sciences and Arts
marc.bravin@hslu.ch

**Melanie Clegg**
University of Lausanne
melanie.clegg@unil.ch

**Reto Hofstetter**
University of St. Gallen
reto.hofstetter@unisg.ch

**Marc Pouly**
Lucerne University of Applied Sciences and Arts
marc.pouly@hslu.ch

**Jonah Berger**
University of Pennsylvania
jberger@wharton.upenn.edu

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These materials have been supplied by the authors to aid in the understanding of their paper.

The AMA is sharing these materials at the request of the authors.

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## Web Appendix A: Pilot Study with Content Creators

We recruited 25 content creators ($M_{age}$ = 29.24, SD = 1.05; 72% female) via a content creator recruiting platform (Reachbird.io) to participate in a survey about social media trends and trend following (sample statistics can be found in Table W1).

As a lead into the study, participants first received the following information regarding trends and dance trends:

> **Topic of this survey: Trends on social media**
> This questionnaire is about content creation and trends on social media. Trends are defined as pieces of content that follow a popular screenplay or choreography, oftentimes designated with a common hashtag. An example could be TikTok videos that follow a similar screenplay (like the cleaning challenge) or music and dance challenges.
> On the next pages, we will ask you a couple of questions about how you adopt trends in your content creation. Filling out the whole survey will take you approx. 5-10 minutes.

Then, participants answered several questions related to trends and trend following. First, they were asked how important they consider trends for their social media practice ("Overall: How important do you consider trends for content creators?"; 1 = not at all important, 7 = very important). Second, they were asked how often they create content following a trend ("How often do you create content based on a trend (i.e., follow a trend) on social media?"; answer options ranging from "never" to "at least once per day"). Third, they indicated in an open-ended question what their main considerations for planning trend-related content would be. Fourth, they indicated whether they would favor following a trend closely or deviate ("If you would decide to follow a trend: Would you stick to this trend as close as possible or aim to deviate from the trend?"; 1 = follow as close as possible (do the trend 100% identically to everyone else), 7 = deviate as much as possible (0% only use the same sound but do something completely different)). Fifth, participants answered two open-ended questions, why and how they would deviate.

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Finally, participants indicated their age, gender, number of followers, most important social media platform, and could disclose their content creator name on the platform they considered most important for them (optional).

Table W1: Sample Statistics for the Pilot Study

| Mean age | Gender | Mean number of followers | Most important platform |
|---|---|---|---|
| 29.24 | 72% female | 159,248 | Instagram: 64%<br>Snapchat: 8%<br>TikTok: 12%<br>YouTube: 16% |

***Results and Discussion***

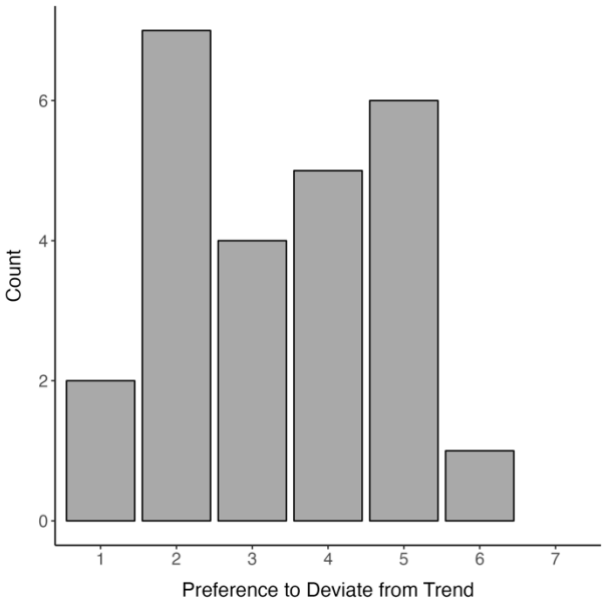
Results imply that content creators consider trends important on social media. Seventy-six percent indicated that following trends is important. The average response was significantly above the scale's mid-point of 4 ($M$ = 5.12, SD = 1.69; $t(24) = 3.11$, $p = .005$). Moreover, 80% indicated that they produced trend-related content at least once per month.

Results also show that content creators are divided whether they favor closely following or deviating from a trend. Fifty-two percent of respondents indicated that they favor following a trend closely, 20% did not indicate a preference (i.e., they selected the mid-point of the scale), and 28% favored deviating from the trend (see Figure W1). This dividedness is also mirrored in responses to the two open-ended questions related to why and how they deviate. Whereas some answered that they "like to stick with the trend" or would "stay as close as possible," others mentioned that they "will try to bring some own variations in the trend" or would "variate with the choreography and use the same sound."

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Figure W1: Distribution of Respondents Favoring Deviating (vs. Not Deviating) From Trends




*Notes:* Scale ranged from 1 = follow as closely as possible (do the trend 100% identically to everyone else) to 7 = deviate as much as possible (0% only use the same sound but do something completely different).

These results point out the relevance of our research topic: First, trend following is considered an important strategy for content creators. Second, for content creators it is not intuitively clear how they should best approach a trend: follow it closely or deviate.

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**Web Appendix B: Determining Video Atypicality**

***Challenges of Automatic Video Analytics and how MUVID Addresses Them***

Videos were projected to account for 82% of internet traffic (Business Insider 2017), presenting both opportunities and challenges for marketing research (Baverstock 2019). Video content is a rich source of information, yet the analysis of video content is not straightforward. Traditional methods, such as manual coding, are often time-consuming and subjective, leading to inconsistencies and potential biases. Automated video analysis may overcome these obstacles, but the implementation of these techniques is not without its challenges.

First, videos are inherently *multimodal* and comprise visual, textual, and audio components. The fusion and synchronization of these modalities is difficult. Therefore, earlier research focused either on the visual information only (e.g., Lu, Xiao, and Ding 2016) or relied on multiple models to separately analyze visual and audio information (e.g., Li, Shi, and Wang 2019; Rajaram and Manchanda 2023; Schwenzow et al. 2021).

Second, videos are inherently *dynamic*, containing a stream of information over time. This complexity has led prior research to analyze video frames independently, using methods from image analytics (e.g., Li, Shi, and Wang 2019). Other research compensates for temporal changes between frames by applying one model for image analytics and another model to combine the extracted information from the individual frame images (e.g., Yang, Zhang, and Zhang 2025).

Third, as video data continues to proliferate, *scalability* becomes a critical issue for video analytics. Marketing researchers have often relied on manual video coding (e.g., Akpinar and Berger 2017; Smith, Fischer, and Yongjian 2012; Teixeira, Picard, and el Kaliouby 2014; Teixeira, Wedel, and Pieters 2010). Manual coding can be useful, but it is

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time-consuming, often subjective, and difficult to scale. Coding ten times more videos, for example, takes ten times longer, and different coders' biases can have strong influences. Moreover, this scalability issue is further compounded by the nature of video annotation itself. Unlike images that can be rapidly evaluated and labeled, video annotation requires careful viewing of the entire video, given the way video context and meaning can evolve. This laborious process forms a significant bottleneck in the creation of large-scale annotated video datasets, posing a substantial challenge to the scalability of video analytics.

Fourth, different applications of video analytics require customized approaches and algorithms. Developing *versatile* pre-trained models capable of addressing a wide range of research questions while maintaining both accuracy and efficiency presents a significant challenge, particularly given the richness of video data. Previous research started to develop approaches that can extract more general video features, such as motion or color features (e.g., Zhou et al. 2021). However, depending on the research questions, more specific, research-related analytics may be necessary.

Fifth, implementing algorithms for a specific research context can be challenging for non-experts. The complexity of algorithms, hardware requirements, and integration with existing systems may hinder widespread adoption. Some researchers have started to open-source their raw code (e.g., Schwenzow et al. 2021), but applying these methods to new research contexts requires expert-level programming skills. For text and images, online converter tools make complex models *accessible* through user-friendly interfaces (e.g., WritingClarityCalculator.com, Warren et al. 2021), but comparable video converter tools are still lacking.

In sum, the surge in video content offers valuable insights for marketing research but poses challenges. Analyzing videos is complex, manual coding is time-consuming, and automated analysis faces challenges due to the *multimodal* and *dynamic* nature of videos.

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Moreover, automated video analysis approaches should be *scalable*, *versatile*, and *accessible*to researchers. As of now, there are no automated video analytics tools available that are specifically tailored to social media research, i.e., tools that use a model pre-trained on social media videos.

MUVID addresses these challenges. First, it is *multimodal*, incorporating both visual and audio information within one model while synchronizing both modalities. Second, it is *dynamic*, accounting for temporal aspects, movements, and object interactions. Third, it is *scalable*, using an unsupervised algorithm to learn features without the need for labeled data. Fourth, it is *versatile*, outputting embeddings that can be used for various downstream tasks using transfer learning. Fifth, it is *accessible*, automatically processing uploaded video data through an interactive online app. Table W2 highlights the key challenges in automated video analytics and compares prior research with our proposed MUVID model.

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Table W2: Characteristics of Video Analytics and How MUVID Addresses Them

| **Challenges of Automated Video Analytics** | **Prior Marketing Research** | **Advantages of MUVID** |
|---|---|---|
| Multimodal | - Focus on only visual information (e.g., Lu et al. 2016)<br>- Relied on multiple models for different modalities (e.g., Li et al. 2019; Rajaram and Manchanda 2023; Schwenzow et al. 2021) | - Incorporates visual and audio information as separate embeddings<br>- Synchronizes both, allowing for cross-modal retrieval |
| Dynamic | - Analyzed single frames without accounting for temporal changes (e.g., Lu et al. 2016; Rajaram and Manchanda 2023; Teixeira et al. 2012)<br>- Additional model to incorporate temporal changes within frames (Yang, Zhang, and Zhang 2025) | - Extends beyond static frames<br>- Accounts for temporal aspects<br>- Accounts for movements and object interactions |
| Scalable | - Relied on manual video coding (e.g., Akpinar and Berger 2017; Smith et al. 2012; Teixeira et al. 2014; 2012)<br>- Used pre-trained models from image analytics (e.g., Rajaram and Manchanda 2023) | - Learns features without labeled data<br>- Unsupervised learning algorithm makes it scalable for processing large-scale video datasets<br>- Embedding representations compress videos while retaining semantic information |
| Versatile | - Predicted a fixed set of research-context dependent features such as the presence of a brand logo or emotions (e.g., Lu et al. 2016; Rajaram and Manchanda 2023) | - Generates embedding representations for various subsequent analyses<br>- Facilitates tasks such as clustering, video categorization, or engagement prediction |
| Accessible | - Source code publicly available but still requires programming knowledge to apply to new data (e.g., Schwenzow et al. 2021) | - Available via an interactive online research app<br>- Less programming knowledge is required to process videos<br>- Open-source model and code |

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### *TikTok Video Dataset: Data Collection Process*

The objective of our data collection process is to obtain a diverse dataset of trending videos from different creators and categories. Collecting a dataset from TikTok is not straightforward, as the recommender algorithm influences what videos are shown to the users and, thus, also to the scraper. On TikTok, users do not exclusively consume content from creators they are following. Instead, TikTok presents the content to the users on their personalized "For You" page as an endless stream of videos (Bhandari and Bimo 2020) that

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was termed *unpredictable* elsewhere (Simpson and Semaan 2021). Instead of typical social network features like profiles with user information, friends lists, and communication tools (Boyd 2010), TikTok focuses on displaying users' video content, and statistics such as followers and likes, and allows commenting on videos. Interaction on TikTok primarily revolves around video content curated by the app's recommendation algorithm (Klug et al. 2021). Given that recommender algorithms usually influence user preferences and trending topics on user platforms (Liu and Cong 2023), gathering a sample based on this recommender algorithm was essential for the topic of trend following.

Therefore, similarly to Tian, Dew, and Iyengar (2024), who restricted their scraping strategy to the "Discovery" page of TikTok, we aim to have a sample of videos that a typical user would likely discover. Typically, there are three ways users interact with the TikTok platform and get exposed to videos: through the recommender system, by searching for videos using hashtags, and by searching for specific creators.

As a result, we follow a multi-step process. First, we construct a list of creators' names, entering all letters from the alphabet and numbers from 0 to 9 into TikTok's search function and selecting a maximum of 100 creators that appeared for each initial character. The follower sizes of these creators range from zero to several million. For all creators on this list, we gather all videos they have uploaded. Second, inspired by the most popular TikTok hashtags and categories (Ceci 2022; Freer 2022), we create a list of hashtags and manually assign them to 14 different categories (see Table W3 for original hashtags and hashtag-to-category mapping for downstream evaluation). We then scrape videos that were found using these hashtags. Finally, we collect videos that were recommended by the TikTok recommender.

We repeat this process over a period of 12 months and end up with a large dataset of 864,294 videos that were uploaded between June 2015 and July 2022. In total, this process

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results in 126,845 different creators with varying numbers of followers (min = 0; max = 144.8 M). Categorizing these creators into nano-, micro-, macro-, and mega-influencers (according to Campbell and Farrell 2020) shows that the majority are nano-influencers (see Figure W2). The dataset further includes descriptive information about the videos, such as hashtags and music played, and content-related statistics, such as the views, likes, comments, and share counts (see Table W4).

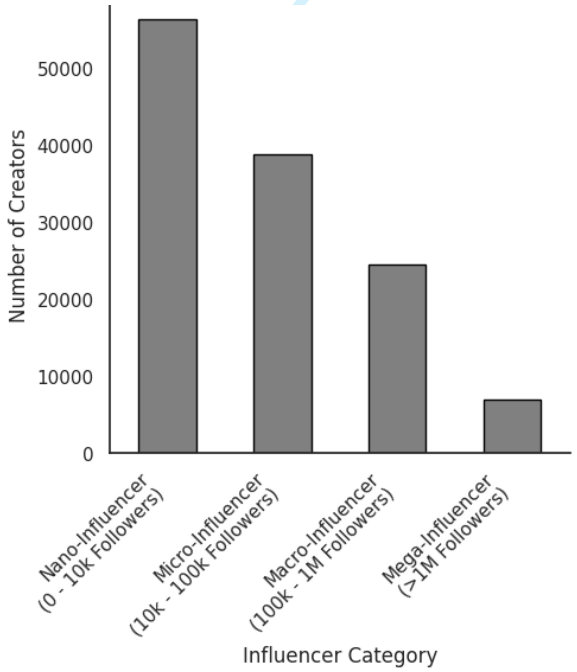
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Table W3: Overview of Hashtags and Hashtag-to-category Mapping Used to Obtain a Labeled Dataset of 14 Categories for Downstream Evaluation

| Category | Sample Hashtags | #Videos |
|---|---|---|
| animal | animal, cat, pets, dog, kitty | 5,615 |
| art | comic, drawing, art, sketch, artist | 1,915 |
| beauty | beauty, skincare, skincareroutine, skincareadvice, beautytips | 3,712 |
| bike | cycle, bike, motorcycle, motorrad, fahrrad | 1,298 |
| car | car, cars, auto, supercar, bmw | 3,963 |
| cooking | food, cooking, cake, recipe, cook | 2,718 |
| dance | dancing, dance, dancinthug, dancingchallenge, badboydance | 110,924 |
| fashion | tiktokfashion, fashion, clothing, cloths, sewing | 2,275 |
| fingerdance | fingerdance, fingertutting, tutting, handdance, fingertuting | 2,228 |
| lifehacks | diy, lifehacks, homereno, renovation, tutorial | 3,567 |
| outdoor | outdoor, outdoors, sun, sunset, sunrise | 3,673 |
| prank | prank, pranks, tiktokprank | 1,795 |
| sport | sports, gym, sporty, sport, workout | 4,757 |
| toddler | baby, toddler, kids, funnytoddler, cutekid | 1,998 |

Figure W2: Categorization of Content Creators in the Dataset




*Notes*: The figure displays the distribution of content creators in the dataset by follower count, illustrating that the majority are nano-influencers.

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Table W4: Overview of Variables in the Scraped Dataset

| Variable Type | Description |
|---|---|
| ID | Unique identifier of the video |
| Date | Date and time the video was posted |
| Crawling Date | Date and time the video has been scraped from the platform |
| Video | Video-file (MP4)<br>Video duration in seconds |
| Audio | Music ID<br>Music Title<br>Music original (whether the audio track is original)<br>Music TikTok (whether the user has selected a music from the TikTok library) |
| Author | Author name<br>Number of followers<br>Number of likes the author received<br>Author Verified (whether author has been verified by TikTok) |
| Hashtags | Hashtags associated with the video |
| Description | Textual description added by the author |
| Duet Video ID | ID of the original video (only available if the video is a duet) |
| Video Stickers | The list of stickers added to the video |
| Video Effects | The list of effects used in the video |
| Warning Information | Optional warning information added by TikTok |
| Metrics | Number of likes (from time of posting to time of scraping)<br>Number of views (from time of posting to time of scraping)<br>Number of comments (from time of posting to time of scraping)<br>Number of shares (from time of posting to time of scraping) |

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### *MUVID Training Procedure*

If no labels are available, self-supervised learning is the method of choice (Le-Khac, Healy, and Smeaton 2020). Among self-supervised methods, contrastive learning is a powerful unsupervised training paradigm that has been successfully applied to image, text, and audio analysis (e.g., Al-Tahan and Mohsenzadeh 2021; Chen et al. 2020; Radford et al. 2021). It involves training a model to distinguish between positive pairs—different augmented views of the same sample—and negative pairs—different samples. This training

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procedure enables the model to learn embeddings that are closer together for positive pairs and farther apart for negative pairs, effectively capturing semantic representations of the data. The training procedure of MUVID is based on a contrastive learning paradigm, which consists of four crucial steps.

First, using the *SimCLR* framework, we generate positive and negative pairs of videos. Second, we use a contrastive loss-function to extract embeddings that best capture similarities and dissimilarities between positive and negative pairs using the NT-Xent loss function (Sohn 2016). The intuition behind this approach is that augmentations of the same video (positive pairs) should be more similar than augmentations of different videos (negative pairs), and extracted embeddings are learned such that they capture this information best. Importantly, the model extracts audio and video data separately and in combination to account for the multimodality of videos. Third, we apply this architecture to train the model on our data. Fourth, we use the model to extract the embeddings separately for video and audio streams. Finally, we evaluate the video performance. Each of these steps is explained in more detail below.

*Model training step 1: Generating positive and negative pairs*

First, inspired by the *SimCLR* framework (Chen et al. 2020) proposed for learning image embeddings, we use augmentation techniques to artificially generate positive and negative pairs. We extend it to video data and define positive pairs as visual and audio samples that belong to the same clip. More specifically, to construct positive pairs, we randomly generate two versions of the same video by applying data augmentation operations in both the visual and audio domains. To generate positive pairs in the visual domain, we apply a combination of spatial and temporal augmentations, such as cropping, resizing, color distortions, and frame sampling (Chen et al. 2020; Qian et al. 2021).

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Building on prior work in audio processing, we apply a variety of spectrogram augmentation techniques, such as time warping, frequency masking, time masking, and temporal sampling, to generate positive samples in the audio domain (Park et al. 2019). Note that we maintain temporal consistency across the visual and audio domains to ensure that the model learns to associate and synchronize information from both modalities effectively. Finally, for negative pairs, we use augmented samples from different video clips.

*Model training step 2: Contrastive loss function*

The second step involves defining a contrastive loss function that facilitates the learning of powerful embedding representations. Again, inspired by the SimCLR framework for learning image representations (Chen et al. 2020), we extend the *normalized temperature-scaled cross-entropy* loss function (NT-Xent; Sohn 2016) to operate on the input of multimodal data. In general, the NT-Xent loss learns embedding representations by maximizing the similarity between positive pairs while minimizing the similarity between negative pairs. Our proposed multimodal NT-Xent loss involves three parts. First, to learn powerful visual embeddings, we compute the NT-Xent loss with the output from the visual network. Second, to learn powerful audio embeddings, we compute the NT-Xent loss with the output from the audio network. Third, to synchronize the embeddings from both modalities, we learn a joint visual-audio embedding space by computing the NT-Xent loss for the combinations of visual and audio features. Specifically, we generate positive and negative pairs for each video for both the visual and the audio stream. We then maximize and minimize the distances between resulting embeddings for these four information streams. Importantly, the model extracts audio and video data separately and in combination to account for the multimodality of videos.

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*Model training step 3: Training*

The third step involves training the model. We implement the proposed MUVID architecture and training procedure using the deep learning framework TensorFlow version 2.9 and train it on a machine equipped with a Quadro RTX 8000 GPU with 48 GB of memory. We train the model using a batch size of 75 pairs for a total of 20 epochs, which took more than one week to complete.

*Model training step 4: Embedding extraction for further processing*

After training the model, we individually extract visual and audio embeddings to facilitate separate analysis of both modalities. Since the model was trained on short clips consisting of 16 frames, we divide the full video into shorter clips and feed each clip to the visual and audio networks to extract the embedding representations. Then, we compute the average visual and audio embedding for each video. These video-level embeddings provide a consolidated representation of the visual and audio modalities for each video, allowing us to use them for further processing. Next, we report performance indicators that test whether the embeddings indeed accurately capture the semantic representations of the video content.

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***More Detailed Information about the MUVID Augmentation Techniques***

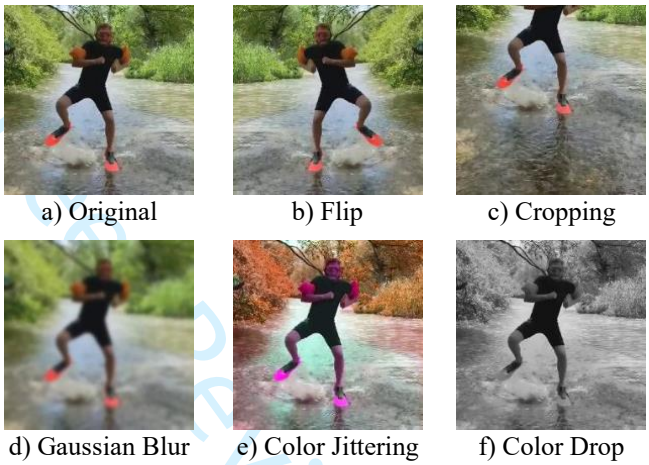
The core of our unsupervised pre-training procedure is the augmentation techniques. We combine temporal and spatial augmentation methods. We implement the following spatial augmentation techniques (see Figure W3).

- *Flip:* random left-right flip of all video frames.
- *Random Cropping:* a region of interest that matches the target size is randomly selected. All frames in the video are then cropped according to this region.
- *Gaussian Blur:* randomly applying Gaussian blurring with a kernel size of 10% of the frame size and a randomly sampled parameter $\sigma \in [.1, 2.0]$

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- *Color Jittering:* randomly adjusting the brightness, contrast, saturation, and hue values of the frames.
- *Color Drop:* randomly converting all frames to grayscale.

Figure W3: Spatial Augmentations to Generate Positive Pairs for Contrastive Learning



a) Original b) Flip c) Cropping

d) Gaussian Blur e) Color Jittering f) Color Drop

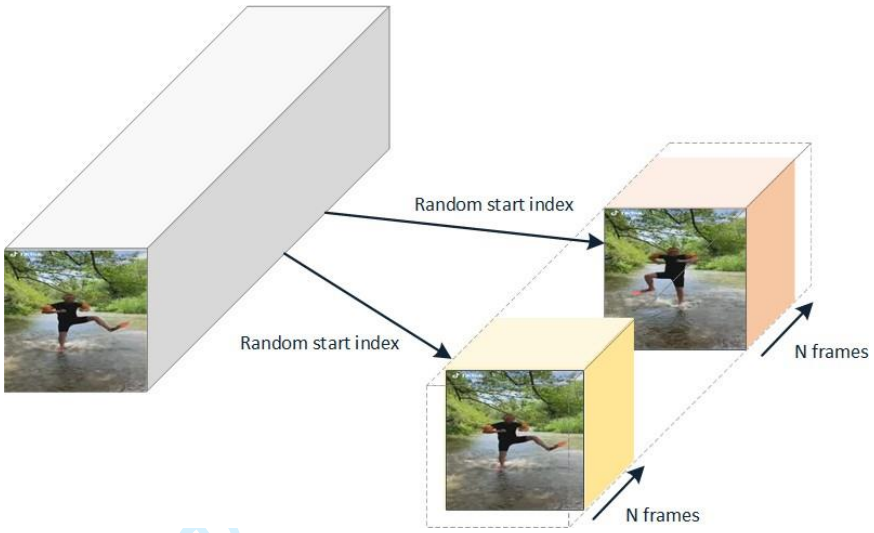
We implement an additional temporal augmentation technique inspired by Qian et al. (2021). For each video $V = (x_1, x_2, \ldots, x_T)$ where $x_i$ represents the i-th frame and T the total number of frames, we start by randomly sampling a start frame. From this randomly selected start frame, we then select the next N subsequent frames to form a short clip.

Since our videos were processed at four frames per second (FPS), there could be instances where the total number of frames T in a video is less than N. In such cases, we repeat the video frames until $T \geq N$. This approach ensures that each clip consistently contains N frames. An illustration of the temporal augmentation is shown in Figure W4.

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Figure W4: Temporal Augmentations Applied During Training Procedure




Since the audio data was transformed into Mel spectrograms before training, traditional audio augmentation techniques that modify the raw audio signal cannot be applied. Instead, the spectrograms were temporally split into several smaller ones, similar to the temporal augmentation for the videos. To further augment the audio information, we applied several spectrogram augmentations (Park et al. 2019; see Figure W5). Given a spectrogram with $\tau$ lengths on the time axis and $\nu$ lengths on the frequency axis, the following augmentation methods were implemented:

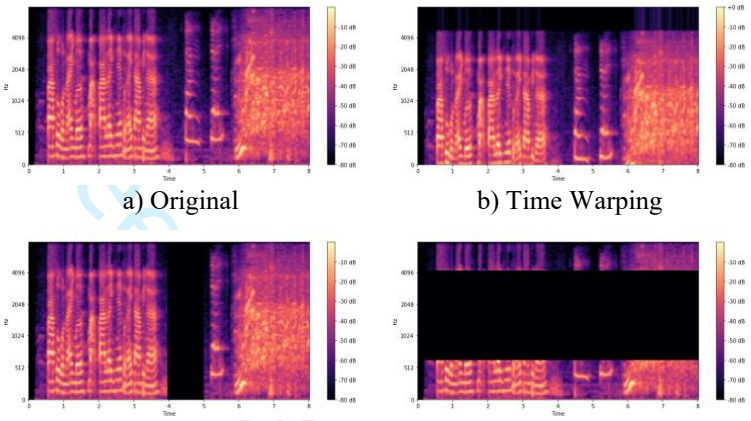
- *Time Warping:* A source point in the time axis is randomly chosen from $[\tau/4, \tau - 4]$. This point is warped by a time distance $w \in [-W_\tau, W_\tau]$, where $W_\tau$ is the time warp parameter and was set to the default value $W_\tau = 80$.
- *Frequency Masking:* This augmentation policy masks $f$ consecutive Mel frequency channels $[f_0, f_0 + f)$, where $f$ is a discrete random variable $\in [0, F]$ and F is the frequency masking parameter. The parameter $f_0$ is chosen from $[0, \nu - f]$. The selected region is replaced by the minimum value. This is repeated $N_f$ times with default parameters $F = 27$ and $N_f = 2$.
- *Time Masking:* t consecutive time steps $[t_0, t_0 + t)$ are selected, where $t \in [0, T]$ is a discrete random variable. The parameter T is the time-masking parameter and $t_0$

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chosen from $[0, \tau - t]$. The selected region is replaced by the minimum value. This is repeated $N_t$ times with default parameters $T = 100$ and $N_t = 2$.

Figure W5: Spectrogram Augmentations Applied to the Audio Information to Generate Positive Samples for Contrastive Learning



a) Original b) Time Warping

c) Frequency Masking d) Time Masking

***Training Procedure***

Once we have generated two different versions of the same video, we feed them into the visual and audio networks, respectively, to obtain two feature representations in each domain. We then maximize the cosine similarity between augmented versions from the same video while minimizing the cosine similarity between augmented versions from different videos. This enables the model to bring positive embedding pairs closer together while pushing negative embedding pairs further apart.

More specifically, we optimize the *normalized temperature-scaled cross-entropy* (NT-Xent) loss function (Sohn 2016) given in Equation W1 that is fed with two embeddings $\boldsymbol{z}_i$ and $\boldsymbol{z}_j$. The function *sim* denotes the cosine similarity, $\tau$ *is* a hyperparameter that scales the similarities and the indicator function $\mathbb{1}_{[k \neq i]} \in \{0, 1\}$ evaluates to 1 if, and only if, $k$ is not equal to $i$.

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$$\mathcal{L}(\boldsymbol{z}_i, \boldsymbol{z}_j) = -\log \frac{\exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_k)/\tau)} \quad \text{(W1)}$$

To train MUVID, we define a multi-modal NT-Xent loss. First, to learn powerful visual embeddings, we compute the loss with the two different visual embeddings. Second, to learn powerful audio embeddings, we compute the loss with the two different audio embeddings. Third, to synchronize the visual and the audio modalities, we learn a joint visual-audio embedding space by computing the loss for the four different combinations of visual and audio embeddings. We implemented MUVID using TensorFlow version 2.9 and trained it on a Quadro RTX 8000 GPU with 48 GB memory. The full model configuration is given in Table W5.

Table W5: Full Configuration of the MUVID Model

| Stage | Configuration | Value |
|---|---|---|
| Visual Network | Model Architecture | R(2+1)D ResNet-18 |
| | Embedding Shape | 512 |
| | Projection Layers | [512, 128] |
| | Number of Frames | 16 |
| | Frame Width | 112 |
| | Frame Height | 112 |
| | Frames per Second (FPS) | 4 |
| | Stride Size | 4 |
| Audio Network | Model Architecture | 2D ResNet-18 |
| | Embedding Shape | 512 |
| | Projection Layers | [512, 128] |
| | Number of Mels | 128 |
| | Number of Units | 31.25 |
| | Sampling Rate | 16 kHz |
| | Hop Length | 512 |
| | Number of FFTs | 2048 |
| Trainer | Max. Number of Epochs | 20 |
| | Batch Size | 75 |
| | Temperature (NT-Xent Loss) | 0.1 |
| | Optimizer | Adam |
| | Adam Beta 1 | 0.9 |
| | Adam Beta 2 | 0.999 |
| | Adam Epsilon | 1e-7 |
| | Learning Rate | 1e-4 |

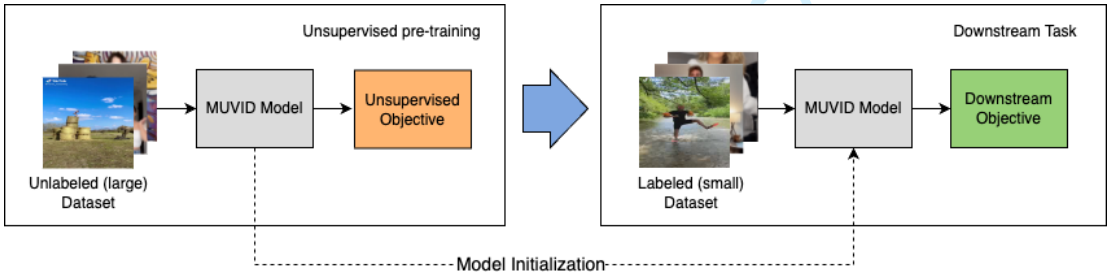
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***Evaluation of MUVID Performance***

Given that MUVID was not trained using labeled data, direct measurement of model performance (such as accuracy in supervised machine learning) is not possible. Consequently, to assess performance, we apply an established computer science methodology called *downstream evaluation* (e.g., Schiappa, Rawat, and Shah 2023; see Figure W6). This method involves defining downstream tasks (e.g., predicting video categories) where labeled data is available and then training a supervised learning model using the generated embeddings from MUVID as input features. Specifically, we employ linear classification and regression heads—simple linear models trained on top of the MUVID embeddings—to classify or predict given outcomes. Linear models are chosen for their simplicity and effectiveness in evaluating the separability and quality of the embedding space. The performance metrics (e.g., classification accuracy or regression error) measured on a holdout test set are interpreted as proxies for the embedding model's quality. Additionally, the model's performance is compared against multiple alternative models (e.g., models used in prior literature) to benchmark MUVID's effectiveness.

Figure W6: Downstream Evaluation Procedure




*Notes:* First, MUVID is pre-trained using an unsupervised training objective (left figure). Second, MUVID is evaluated by defining a set of downstream tasks where labeled data is available (right figure). The MUVID model is initialized using the pre-trained weights and it is then used in a supervised learning setting where the performance of the model is assessed on the downstream tasks.

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*Downstream tasks for quantitative model evaluation*

We use four downstream evaluation tasks. For each, we generate a labeled subset from the original dataset. We first test whether the embedding representations can accurately predict video categories. As most videos in the dataset have hashtags, we utilize them as category labels. Inspired by the most popular TikTok hashtags and categories (Ceci 2022; Freer 2022), we manually define a hashtag-to-category mapping table, assigning videos to one of 14 different categories (e.g., dance, cars, or cooking; see Table W3). We sample 1,000 videos from each category, resulting in a balanced dataset of 14,000 labeled videos.

The second downstream task assesses whether the embeddings can accurately predict the music titles associated with the videos. To create a balanced labeled dataset, we select videos that use one of the top 50 most frequently used music titles and sample 400 videos for each, resulting in 20,000 videos.

The third downstream task evaluates whether the MUVID embeddings can accurately predict engagement. Due to the highly skewed distribution of the engagement measure, we apply a logarithmic transformation before using it as a target variable.

The fourth downstream task introduces a retrieval task to evaluate the ability of the embeddings to identify atypical videos, functioning as an outlier detection benchmark. For this, we select the top five trends from our trend-dataset. For each trend, we create a test set by randomly sampling 180 videos from that trend and adding 20 random videos from a category other than 'dance' to serve as outliers. This composition results in a test set of 200 videos with a 10% contamination rate. Then, for each of the 200 videos in the test set, we calculate its atypicality score as the average cosine distance from its embedding to all other videos in the dataset. By sorting the videos in descending order of this score, we evaluate the model's performance using the Precision@k metric per trend and then compute the average precision and the standard deviation. This metric measures the proportion of the top k videos

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in the sorted list that are correctly identified as outliers and thus essentially not part of the trend.

*Comparison baselines for model performance*

To estimate model performance, we compare its prediction and classification accuracy for the downstream tasks against multiple alternative models. First, we compare it against pre-trained models for video and audio analytics (see Table W6). Note that for a fair comparison, we keep the frame and audio sampling rates consistent across all models and baselines.

Table W6: Overview of the Benchmark Models to Compare Against the MUVID Model

| Name | Modality | Description | References |
|---|---|---|---|
| ResNet-18 | Visual | A 2D ResNet-18 CNN model pre-trained on the ImageNet dataset. We extracted embeddings for each frame of an input video and computed the average across all frames to obtain video-level embeddings. A similar architecture has been used in previous work in marketing (e.g., Rajaram and Manchanda 2023; Yang, Zhang, and Zhang 2025). | He et al. 2016 |
| R2Plus1D | Visual | An R(2+1)D ResNet-18 model identical to the Visual-Network of MUVID. Pre-trained on Kinetics-400 Action Recognition dataset (Kay et al. 2017). | Tran et al. 2018 |
| VideoMAE | Visual | A video transformer model that has been trained in an autoencoder paradigm | Tong et al. 2022 |
| AST | Audio | A pre-trained audio spectrogram transformer that transforms audio signals into embedding representations | Gong, Chung, and Glass 2021 |
| YAMNet | Audio | A pre-trained neural network for sound event detection based on the MobileNet v1 architecture | Howard et al. 2017 |

Second, we compare it against the MUVID audio-only embeddings and visual-only embeddings, respectively. According to best practices in machine learning, each dataset (category, music, engagement) was split into a training and test set (80%/20% split). Using the training sets, we train separate linear models for each task and embedding type and evaluate their performance on the test set. We intentionally choose linear models based on the assumption that if the embeddings effectively capture the semantic representations of the

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videos, such patterns should be identifiable by a linear model (Schiappa, Rawat, and Shah 2023).

*Model evaluation results*

MUVID outperforms competing baselines in the classification tasks (see Table W7). It can classify the category and predict engagement more accurately than all competing approaches based on repurposing pre-trained video or image models. Only in the music title classification task does MUVID not outperform existing music classification models.

Table W8 shows the results of the retrieval task. Again, the MUVID model outperforms competing benchmarks, except the audio models. However, this makes sense since all videos from the trend share the same music title, and thus an audio model should naturally perform better than a visual-only model.

Overall, this evidence suggests that for the processing of short social media videos, MUVID is the highest-performing alternative. Only highly specialized models have advantages in music classification, which is, however, not the main focus of our analysis.

Table W7: MUVID Evaluation Benchmarks for Classification Downstream Tasks

| | **Category** | | **Music** | | **Engagement** | |
|---|---|---|---|---|---|---|
| **Embeddings** | **Acc.** | **$F_1$** | **Acc.** | **$F_1$** | **$R^2$** | **MAE** |
| MUVID Visual + Audio | 52.46 | 51.80 | 85.08 | 86.06 | .241 | .515 |
| MUVID Visual | 50.75 | 49.73 | 26.70 | 24.29 | .212 | .524 |
| MUVID Audio | 24.96 | 22.59 | 84.18 | 85.13 | .146 | .552 |
| ResNet-18 | 14.25 | 7.92 | 3.58 | .89 | .055 | .583 |
| R2Plus1D | 35.21 | 28.79 | 7.45 | 3.90 | .125 | .560 |
| VideoMAE | 34.61 | 32.20 | 8.92 | 5.66 | .177 | .542 |
| AST | 29.71 | 27.80 | 89.72 | 90.18 | .147 | .555 |
| YAMNet | 30.07 | 28.59 | 88.48 | 88.81 | .111 | .568 |

*Notes:* Acc. = Accuracy (percentage), $F_1$ = macro $F_1$ score (percentage), MAE = Mean Absolute Error. MUVID Visual indicates the visual embeddings from the visual network. MUVID Audio indicates the audio embeddings from the audio network. MUVID Visual + Audio indicates the full MUVID embeddings by concatenating both visual and audio embeddings.

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Table W8: MUVID Evaluation Benchmarks for the Retrieval Task

| **K** | **5** | | **10** | | **20** | |
|---|---|---|---|---|---|---|
| **Embeddings** | **Precision** | **Std.** | **Precision** | **Std.** | **Precision** | **Std.** |
| MUVID Visual + Audio | 96 | 8.94 | 98 | 4.47 | 94 | 8.22 |
| MUVID Visual | 96 | 8.94 | 96 | 5.48 | 79 | 10.84 |
| MUVID Audio | 96 | 8.94 | 96 | 8.94 | 94 | 5.48 |
| ResNet-18 | 0 | 0 | 8 | 8.37 | 10 | 7.90 |
| R2Plus1D | 20 | 0 | 16 | 8.94 | 19 | 8.22 |
| VideoMAE | 60 | 0 | 54 | 5.48 | 42 | 7.58 |
| AST | 100 | 0 | 100 | 0 | 97 | 4.47 |
| YAMNet | 100 | 0 | 98 | 4.47 | 95 | 5.00 |

*Notes:* Precision = Precision at K in percentage. Std = Standard Deviation across all Trends. MUVID Audio indicates the audio embeddings from the audio network. MUVID Visual + Audio indicates the full MUVID embeddings by concatenating both visual and audio embeddings.

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## Web Appendix C: Evidence from Secondary Data

### *TikTok Trend Dataset*

Table W9 provides a comprehensive overview of the variables utilized in our secondary data analysis. It details the operationalization of our focal dependent variable (Engagement), our primary independent variable derived from the MUVID visual embeddings (Overall Atypicality), and the various control measures. These controls are systematically categorized into video-level characteristics (including machine-learning-derived quality scores and text atypicality), trend-level exposure metrics, and creator-level attributes, ensuring a rigorous accounting of potential confounding factors in our empirical models.

Table W9: Variable Overview of the Secondary Data

| Variable | Description |
|---|---|
| ***Dependent Variable*** | |
| Engagement | Likes per View measured at the time of data collection |
| ***Independent Variables*** | |
| Overall Atypicality | Distance of focal video to all earlier videos from the same trend (computed based on cosine distance of MUVID embeddings) |
| Text Atypicality | Distance of text in focal video to text of all earlier videos from the same trend (computed based on the cosine distance to text embeddings) |
| ***Video Characteristics*** | |
| Time on Platform | Number of days the video was online before it was scraped |
| Dance Quality | Quality of the dance (computed based on supervised training approach) |
| Visual Quality | Overall visual quality of video (computed based on supervised training approach) |
| ***Trend Characteristics*** | |
| Trend Exposure | The number of videos that were posted for a trend (computed based on the sum of videos for the assigned music IDs, collected directly from TikTok) |
| ***Creator Characteristics*** | |
| Creator Followers | Number of followers |
| Creator Verified | Creator verified (yes / no dummy) |
| Creator Num Videos | Total number of posts on the creator profile |

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*Dance classifier*

To develop a dance classifier for filtering TikTok videos, we first assemble a labeled dataset based on dance-related hashtags (see Table W3). Specifically, videos tagged with these dance-associated hashtags were labeled as "dance," while those without these hashtags were labeled as "non-dance." We then train a multi-layer perceptron (MLP) with two hidden layers using this labeled dataset to distinguish between dance and non-dance videos.

For model validation, we split the dataset into 80% for training and 20% for testing. The trained classifier achieves an accuracy of 96.26%, indicating its effectiveness in identifying dance content. Utilizing this dance classifier, we process the full representative TikTok dataset and curate a dance dataset comprising 97,872 videos.

*Descriptives of the trend dataset*

Table W10 provides the summary statistics for our final trend dataset, which encompasses 1,548 distinct trends and content from 25,611 unique creators.

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Table W10: Descriptives of the Trend Dataset

| | N | Mean | Median | SD | Min | Max |
|---|---|---|---|---|---|---|
| Videos per Trend | 1,548 trends | 65.55 | 33 | 93.38 | 10 | 1,885 |
| Videos per Creator | 25,611 creators | 3.96 | 2 | 7.23 | 1 | 403 |
| Likes | - | 87,364.1 | 603 | 507,480.4 | 0 | 25,200,000 |
| Comments | - | 752.62 | 11 | 7,131.26 | 0 | 675,600 |
| Views | - | 746,759.1 | 7,371 | 4,382,432 | 0 | 248,100,000 |
| Shares | - | 2,247.42 | 4 | 231,966.6 | 0 | 52,300,000 |
| Overall Atypicality (MUVID) | - | 0.4 | 0.4 | 0.08 | 0.04 | 0.75 |
| Text Atypicality | - | 0.68 | 0.68 | 0.15 | 0.03 | 1.04 |
| Num. Days Online | - | 836.04 | 635 | 616.34 | 1 | 2,087 |
| Dance Quality | - | 4.46 | 4.43 | 0.41 | 2.25 | 5.82 |
| Visual Quality | - | 4.72 | 4.73 | 0.22 | 2.89 | 5.63 |
| Creator Followers | - | 1,173,331 | 34,100 | 4,934,843 | 0 | 144,800,000 |
| Creator Verified | - | 0.12 | 0 | 0.33 | 0 | 1 |
| Creator Num Videos | - | 505.46 | 250 | 796.76 | 1 | 34,700 |
| Trend Exposure | - | 7,268,524 | 1,928,481 | 13,337,287 | 20 | 77,775,283 |

*Notes:* All values represent raw, unstandardized values. Note that overall atypicality and text atypicality have different underlying scales. N = number of observations; SD = standard deviation.

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### *Quality Control Variables*

To control for the visual quality of the videos and dance quality of the performer in the videos, we compute control variables based on human labels. To this end, five hundred thirty-four Mechanical Turk workers ($M_{age}$ = 36.4; $SD_{age}$ = 11.4; 50% female) manually labeled 1,878 videos from our dataset. The relevant labels are "overall aesthetic visual quality" and "overall dance quality" (1 = very low; 7 = very high). The workers first rate five test videos and then 20 target videos. Each video is labeled by at least three raters.

After collecting the ratings, we calculate the average rating for each video to serve as the target variable for our prediction tasks. Utilizing the MUVID visual embeddings as feature inputs, we evaluate several prediction models by conducting a comprehensive hyperparameter grid search. To establish a performance benchmark, we introduce a dummy baseline that consistently predicts the median rating for comparison purposes.

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We employ five-fold cross-validation to ensure the robustness and reliability of our model evaluations. The performance metrics for each model across both dance quality and visual quality prediction tasks are summarized in Table W11. Among the evaluated models, the random forest algorithm achieves the highest accuracy and consistency in predicting both dance quality and visual quality, outperforming all other models.

Based on these results, we select the random forest model for both prediction tasks. We then apply these optimized random forest models to our full trend dataset, predicting the visual and dance quality scores for each video.

Table W11: The Results of the Different Models for Predicting Visual and Dance Quality

| | Dance Quality | | Visual Quality | |
|---|---|---|---|---|
| **Model** | **MAE** | **$R^2$** | **MAE** | **$R^2$** |
| Random Forest | .714 | .143 | .650 | .042 |
| Gradient Boosting | .720 | .137 | .651 | .038 |
| k-Nearest Neighbors | .719 | .127 | .655 | .023 |
| Decision Tree | .755 | .048 | .658 | .020 |
| Support Vector Regression | .781 | .000 | .667 | -.010 |
| Dummy Baseline (Median) | .773 | -.027 | .662 | -.006 |
| Multi-Layer Perceptron | .870 | -.257 | .807 | -.500 |
| Linear Regression | .864 | -.282 | .795 | -.451 |

*Notes:* Results of the different prediction models using 5-fold cross-validation. As a dummy baseline model, we returned the median value. MAE = mean absolute error; $R^2$ = coefficient of determination.

### *Empirical Challenges and Robustness Checks*

Table W12 provides an overview of the potential empirical challenges associated with our sample construction, model specification, and identification strategy, alongside the approaches to address each challenge. Details of the specific robustness checks are described in the subsequent sections.

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Table W12: Overview of Empirical Challenges and How They Are Addressed

| Domain | Potential Empirical Challenge | Approach to Resolve | Reported in |
|---|---|---|---|
| *Sample Construction and Assumptions* | | | |
| Trend definition | Trend definition | - Robustness checks across different definitions of trends in our sample (different thresholds of trend sizes: 10, 20, 50, 100 videos) | Web Appendix C |
| Familiarity | Users must be familiar with the trend to spot atypicality | - Ran an auxiliary survey among TikTok users asking about their familiarity with trends<br>- Interaction with trend exposure shows the atypicality effect is increased when trend exposure is higher (and users are more familiar with trends) | Study 1; Web Appendix D |
| Audience | Systematic differences between audiences of typical and atypical videos. | - Confirming similar user types across audiences based on user comment text analysis. | Web Appendix C |
| *Model Specification* | | | |
| Measurement | Atypicality operationalization | - Confirming MUVID performance in qualitative verification and multiple downstream tasks against competing models | Web Appendix B |
| Modeling | Atypicality of videos that are posted closer to the focal video may be more influential | - "Moving window" analysis, such that videos in closer temporal proximity receive a higher weight | Web Appendix C |
| Modeling | Engagement must occur relatively early after posting | - Engagement measure uses likes conditional on view<br>- Investigating the timing of most engagement using comment timestamps<br>- Suggested by prior research and industry reports (Hartmann et al. 2021; Graffius 2025) | Study 1; Web Appendix C |
| Modeling | Distributional assumptions of the main aggregated logit model | - Estimating alternative modeling specifications (Poisson regression with view offset and standard OLS) | Web Appendix C |
| *Identification* | | | |
| Endogeneity | Results may be driven by unobserved creator variables | - Gaussian copula to check for endogeneity<br>- Ruling out creator learning | Study 1; Web Appendix C |
| *Other analysis* | | | |
| Video quality | Atypical videos could systematically differ in terms of quality | - Controlling for dance and visual quality | Study 1 |
| Selection bias | Creators' self-selection in producing atypical videos | - Using propensity score matching to address potential self-selection | Web Appendix C |
| Text elements | Text atypicality may drive the effects | - Controlling for the atypicality of text elements | Web Appendix C |
| Early trend quality | Early video quality could influence both the propensity to post atypical videos and engagement later in the trend | - Inclusion of trend-fixed effects | Study 1, Web Appendix C |

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*Propensity score matching*

One potential issue with the correlational data is that creators were not randomly assigned to perform either an atypical or typical interpretation of the trend, and creators may therefore differ systematically between content types. Propensity score matching is an established technique to account for systematic differences within a non-randomly drawn sample by creating artificial control and treatment groups based on observable variables (Garnefeld et al. 2013; Rosenbaum and Rubin 1983).

First, in a binary logistic regression, we calculate each posted video's propensity to be atypical, incorporating trend-fixed effects, time-fixed effects, and the six key control variables related to video quality and creators. Ensuring that variables like dance and visual quality are balanced between the treatment (atypical) and control (typical) group is vital, as these factors can directly affect engagement and, if unbalanced, could lead to an overestimation of the atypicality effect due to inherent differences in content quality. Additionally, balancing videos based on creator popularity (number of followers and verified status) and productivity (number of videos) may be relevant, as any imbalance could result in falsely attributing engagement boosts to content atypicality rather than the influence of well-established creators. Atypicality is determined using a median split of our atypicality measure within each trend.

Second, we link each video in the treatment group (being atypical) to a video from the control group (being typical), by applying a 1:1 nearest neighbor matching procedure without replacement. We only match pairs if the two propensity scores fall within a prespecified tolerance zone (caliper) of .2. Matching results in a dataset of 81,485 videos. The percentage reductions in the differences of the continuous and binary variables between high and low atypicality groups revealed an average score of 32.05% across all predictors, indicating a substantial decrease in the potential for self-selection bias (Eggert, Steinhoff, and Witte

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2019). Next, we compute standardized differences in means before and after matching, revealing a maximum adjusted difference of .038, far below the recommended value of .25 (Boichuk et al. 2019), indicating that the matching process effectively balanced the treatment and control groups.

Because our matching procedure pairs videos across different creators with similar observable profiles to address self-selection bias, we estimate the treatment effect on the matched sample without creator-fixed effects. This approach provides methodological triangulation to our main within-creator specification, demonstrating that the atypicality premium holds even when comparing observationally equivalent videos *between* different creators.

Regression analyses for both unmatched ($\beta = .043$; $SE = .013$; $t = 3.20$; $p < .01$) and matched ($\beta = .037$; $SE = .013$; $t = 2.78$; $p < .01$) data reveal that the results remain consistent with our initial analyses. This suggests that the relationship between atypicality and engagement is robust and that the observed differences between dance quality, visual quality, creators' followers, verified, and number of videos do not alter the result. Tables W13, W14, and W15 show the results of the propensity score matching procedure.

Table W13: Propensity for Atypicality

| DV | Atypical (0/1) |
|---|---|
| Time on Platform | -.060* (.027) |
| Dance Quality | -.251*** (.034) |
| Visual Quality | .149*** (.027) |
| Creator Followers | -.059*** (.012) |
| Creator Verified | -.056 (.034) |
| Creator Num Videos | .070*** (.013) |
| Trend-fixed Effects | Included |
| Time-fixed Effects | Included |
| Observations | 85,869 |
| AIC | 120,686 |
| BIC | 134,998 |

*Notes:* Standard errors in parentheses. Significance codes: ***: < .001, **: < .01, *: < .05. All continuous variables were z-normalized to ensure that they are on similar scales. Unstandardized results do not differ in sign or significance. DV = dependent variable; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion.



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Table W14: Balance Measures Before and After Propensity Score Matching

| **Variable** | **Difference Unadjusted** | **Difference Adjusted** | **Absolute Percentage Reduction** |
|---|---|---|---|
| Time on Platform | .014 | -.012 | 13.41% |
| Dance Quality | -.091 | -.038 | 57.90% |
| Visual Quality | .007 | .005 | 27.33% |
| Creator Followers | -.039 | -.021 | 45.57% |
| Creator Verified | -.006 | -.003 | 42.03% |
| Creator Num Videos | .043 | .018 | 59.38% |
| Average | | | 32.05% |

*Notes:* The objective is to balance the covariates (variables) between the treated and control groups, making them more comparable. After matching, the differences between the groups for each covariate should be minimized, ideally approaching zero. We see that in absolute terms, we achieve a reduction in all variables.

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Table W15: Results Are Robust After Applying Propensity Score Matching

| DV | Engagement | |
|---|---|---|
| Model | Before Matching | After Matching |
| **Atypical (0/1)** | **.043** (.013)** | **.037** (.013)** |
| Time on Platform | -.071** (.027) | -.072** (.028) |
| Dance Quality | -.004 (.011) | .007 (.010) |
| Visual Quality | -.021** (.008) | -.028*** (.008) |
| Creator Followers | -.005 (.003) | -.003 (.003) |
| Creator Verified | .091*** (.018) | .091*** (.019) |
| Creator Num Videos | .038*** (.006) | .042*** (.006) |
| Trend-fixed Effects | Included | Included |
| Time-fixed Effects | Included | Included |
| Observations | 85,869 | 81,485 |
| AIC | 765,495,150 | 712,217,075 |
| BIC | 765,509,472 | 712,231,242 |

*Notes:* Standard errors in parentheses. Significance codes: ***: < .001, **: < .01, *: < .05. All continuous variables were z-normalized to ensure that they are on similar scales. Unstandardized results do not differ in sign or significance. The results show that the effect of being atypical holds even after applying propensity score matching. DV = dependent variable; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion.

*Gaussian copula*

To further account for unobserved heterogeneity in creators, we introduce a copula term into the regression equation. The Gaussian copula can account for the endogenous regressor of atypicality without the need for an explicit instrument (Park and Gupta 2012). We verify the empirical requirement that the endogenous regressor be continuous and non-normal (Becker, Proksch, and Ringle 2022). Although skewness is lower than .77 (in our

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24), our sample size is larger than 2,000 observations, and the Cramér–von Mises statistic is higher than 21.70 ($p < .001$). Therefore, we can assume a copula term with a power of at least 80% (Liengaard et al. 2025; Park and Gupta 2012).

After checking these pre-conditions, we add a Gaussian Copula to our model to account for the correlation between atypicality and the error term such that: $GC = \Phi[H(\text{atypicality})]$, where $\Phi$ is the inverse of the normal distribution function, and $H(\text{atypicality})$ represents the empirical distribution function of atypicality.

We then include this GC term as an additional control variable in our fully specified main model (including all time-varying controls and fixed effects). The copula term is statistically non-significant (GC = .023, $p = .44$), meaning there is no evidence of severe endogeneity biasing our estimates. Following standard econometric practice (Cascio Rizzo et al. 2024; Wlömert and Papies 2019), because the copula control function reveals no remaining endogeneity, we do not include it in our primary specification. This check provides additional confidence that our main creator-fixed effects model is unlikely to be biased by severe endogeneity.

*Ruling out creator learning*

Given that we have a cross-sectional dataset, we cannot fully exclude that talented creators "learn" over time that atypicality pays at some point, receive more engagement, and then produce more atypical videos over time, which may lead to reverse causality and inflate the observed atypicality effect. To rule out this possibility, we test whether the chronological posting order within each creator's sequence of uploads (i.e., the n-th video posted by that creator) has an impact on atypicality. More specifically, we estimate the following OLS regression model:

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$$Atypicality = \beta_0 + \beta_1(posting\ order) + \sum_{i=2}^{N} \beta_i(\text{Control Variables}_i) + \text{Trend FE} + \text{Creator FE} + \text{Time FE} \quad \text{(W2)}$$

We find no significant correlation between posting order and atypicality ($\beta$ = -.0005; *SE* = .0003; $t$ = -1.57; $p$ = .117), which indicates no systematic changes in terms of produced atypicality. Accordingly, creators do not "learn" atypicality or systematically change their atypicality over time.

*Ruling out viewer population shift*

To address the concern that atypical videos might attract a different population of viewers, we conduct a linguistic analysis of comment language. Our premise is that a significant shift in the underlying viewer population would be reflected in the language style of the commenters. This approach is grounded in prior research, which demonstrates that linguistic style can serve as a reliable proxy for audience characteristics (Berger et al. 2020; Han et al. 2023; Kern et al. 2016; Mahajan et al. 2022; Pennebaker and King 1999).

For each of the top 10 trends identified in Study 1, we create two distinct corpora of comments. We randomly sample 100 typical and 100 atypical videos per trend, then collect all associated comments from these 2,000 videos. This results in one large corpus of comments from typical videos and another from atypical videos.

We then use the Linguistic Inquiry and Word Count (LIWC) tool (Pennebaker, Francis, and Booth 2001) to calculate a single, overall Language Style Matching (LSM) score. This score quantifies the degree of linguistic similarity between the two aggregated corpora, with values ranging from 0 to 1, where higher values indicate greater overlap. The resulting LSM score is .92, which is an exceptionally high value at the upper end of the

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typical .75–.95conversational LSM scores. For context, an LSM score of .82 is already considered a very high match (Lie et al. 2024).

This result provides strong evidence of an exceptional linguistic overlap in the language used by commenters on both typical and atypical videos. This suggests that, at least among the segment of viewers engaged enough to leave comments, there is no substantial difference in the type of audience being drawn to atypical versus typical content.

*Controlling for text*

On TikTok, users can add text directly to their videos by adding stickers or add a brief description beneath their videos. To control for the possibility that written information may influence our results, we compute an additional text atypicality control measure. After merging all texts associated with the videos (descriptions, hashtags, sticker texts), we create a text embedding for each video using a pre-trained *Sentence Transformer* (Reimers and Gurevych 2019). We then determine text atypicality by calculating the cosine distances between the text embeddings of a video and those of previously posted videos within the same trend.

Finally, we add this new text atypicality measure as an additional control in our model from Study 1 (including the same controls and fixed effects). The relationship between atypicality and engagement remains significant (Table W16).

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Table W16: Results Are Robust When Controlling for Text Atypicality

| DV | Engagement |
|---|---|
| **Video Atypicality** | **.025*** (.007)** |
| Text Atypicality | .019* (.009) |
| Controls | Included |
| Trend-fixed Effects | Included |
| Creator-fixed Effects | Included |
| Time-fixed Effects | Included |
| Observations | 64,893 |
| Pseudo $R^2$ | .995 |
| AIC | 186,092,343 |
| BIC | 186,205,759 |

*Notes:* Standard errors in parentheses. Significance codes: ***: < .001, **: < .01, *: < .05. All continuous variables were z-normalized to ensure that they are on similar scales. Unstandardized results do not differ in sign or significance. We included all controls and fixed effects as in the main model. DV = dependent variable; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion.

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*Alternative atypicality operationalizations and trend definitions*

One could wonder whether the results are driven by how atypicality is operationalized. Perceived atypicality may only be driven by the first set of videos in each trend, for example. To test this possibility, we compute the similarity to only the first ten videos of each trend. Conversely, maybe perceived atypicality is influenced by all the videos within the trend, regardless of when they were posted. To address this, we compute the distances to all videos within the trend.

We also consider the temporal influence of videos on atypicality. Our initial measure treats all videos posted earlier equally, but in an additional check, we implemented a temporal discounting mechanism using exponential decay ($\lambda = .1$) to give more weight to recent videos. This "moving window" approach means older videos have less influence on atypicality.

account for potential baseline differences in "atypicality" across trends and focus on how much a video deviates from the norm within its own trend context. Across all alternative atypicality operationalizations, however, atypicality is positively related to engagement (see Table W17).

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Table W17: Results Are Robust to Different Atypicality Operationalizations

| DV | Engagement | | | |
|---|---|---|---|---|
| Atypicality Operationalization | Seeding 10 | All | Moving Window | Normalized by Trends |
| **Atypicality** | **.016* (.006)** | **.024*** (.006)** | **.020* (.006)** | **.018*** (.006)** |
| Controls | Included | Included | Included | Included |
| Trend-fixed Effects | Included | Included | Included | Included |
| Creator-fixed Effects | Included | Included | Included | Included |
| Time-fixed Effects | Included | Included | Included | Included |
| Observations | 74,210 | 88,576 | 74,225 | 74,225 |
| Pseudo $R^2$ | .995 | .995 | .995 | .995 |
| AIC | 194,308,726 | 255,005,543 | 193,946,895 | 193,983,521 |
| BIC | 194,435,307 | 255,144,210 | 194,073,498 | 194,110,096 |

*Notes:* Standard errors in parentheses. Significance codes: ***: < .001, **: < .01, *: < .05. We included all controls and fixed effects as in the main model. We vary the atypicality operationalization in each column. Seeding 10 = only use the first 10 videos of each trend as a comparison. All = use all videos of the trends to compute the atypicality measure. Moving Window = implements a temporal discounting mechanism using exponential decay ($\lambda = .1$) to give more weight to recent videos, meaning older videos have less influence on atypicality. Normalize by Trends = atypicality measures are standardized within each trend to account for trend-specific baseline differences, focusing on deviations from the norm within a trend. DV = dependent variable; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion.

Further, results may be influenced by the threshold for creating trends. Trends were initially defined as a music title with at least 10 videos, but to test robustness, we explore trends with at least 20, 50, or 100 videos. Despite the reduction in sample size, results remain comparable or become even more pronounced (see Table W18).

Table W18: Results Are Robust When Varying the Minimum Number of Videos per Trend

| DV | Engagement | | |
|---|---|---|---|
| Min. Videos per Trend | 20 | 50 | 100 |
| **Atypicality** | **.025*** (.007)** | **.026*** (.007)** | **.033*** (.008)** |
| Controls | Included | Included | Included |
| Trend-fixed Effects | Included | Included | Included |
| Creator-fixed Effects | Included | Included | Included |
| Time-fixed Effects | Included | Included | Included |
| Observations | 72,420 | 60,743 | 45,360 |
| Pseudo $R^2$ | .995 | .995 | .995 |
| AIC | 192,412,640 | 169,959,923 | 128,280,272 |
| BIC | 192,534,448 | 170,065,265 | 128,365,682 |

*Notes:* Standard errors in parentheses. Significance codes: ***: < .001, **: < .01, *: < .05. We included all controls and fixed effects as in the main model. We vary the minimum number of videos per trend in each column. DV = dependent variable; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion.

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*Alternative modeling approaches*

To ensure our findings are not driven by the distributional assumptions of the aggregated logit model, we estimate two alternative specifications: a Poisson regression and an ordinary least squares regression.

Because the raw count of likes is mechanically driven by how many times a video is viewed, we must account for differential exposure. In a Poisson framework, this is achieved by including the natural logarithm of views as an offset variable. By using this offset, the model explicitly transforms the dependent variable from raw counts into an engagement rate (likes per view). This approach natively adjusts for the varying reach of each video, functionally achieving the same goal as the aggregated logit model but through a different statistical mechanism.

Matching the fully specified main model in the manuscript, we estimate the effect of trend atypicality on the count of likes (with the log-views offset) alongside all time-varying video- and creator-level control variables, trend-, time-, and creator-fixed effects, with standard errors clustered at the trend level.

Because the Poisson model uses a log-link function, exponentiating the coefficients yields Incidence Rate Ratios (IRRs), representing the relative percentage change in the expected rate of likes for a one-unit increase in atypicality. Atypicality remains positive and significant ($\beta = .021$, $SE = .006$, $t = 3.67$, $p < .001$; Table W19, Model 1). We further estimate a standard Ordinary Least Squares (OLS) regression. While our main aggregated logit model natively accounts for varying exposure and ensures predictions remain bounded between 0 and 1, OLS provides a highly transparent, baseline linear approximation of the effect.

In the OLS specification, we model the raw engagement rate (likes divided by views) directly as the dependent variable. Matching our main, fully specified model, we include all

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time-varying video- and creator-level controls, alongside trend-, time-, and creator-fixed effects, with standard errors clustered at the trend level.

Atypicality remains positive and statistically significant ($\beta$ = .001, $SE$ = .0004, t = 2.83, $p$ < .01; Table W19, Model 2). Because this is a linear-probability approximation, the coefficient indicates an absolute increase in the engagement rate.

These results confirm the robustness of our core finding, that atypicality is positively related to engagement, across different modeling approaches.

Table W19: Results Are Robust to Different Modeling Approaches

| DV | Engagement | |
|---|---|---|
| Model | (1) | (2) |
| **Atypicality** | **.021*** (.006)** | **.001* (.004)** |
| Time on Platform | -.003 (.027) | -.009*** (.001) |
| Dance Quality | -.009 (.008) | -.001* (.0004) |
| Visual Quality | -.010 (.006) | .0001 (.0004) |
| Creator Followers | .003 (.004) | -.001 (.001) |
| Creator Verified | -.001 (.047) | -.001 (.006) |
| Creator Num Videos | -.009 (.023) | .001 (.002) |
| Trend-fixed Effects | Included | Included |
| Creator-fixed Effects | Included | Included |
| Time-fixed Effects | Included | Included |
| Observations | 74,225 | 74,225 |
| Pseudo $R^2$ | .995 | - |
| $R^2$ | - | .782 |
| Adjusted $R^2$ | - | .733 |
| AIC | 170,214,998 | -211,475 |
| BIC | 170,341,601 | -84,872 |

*Notes*: Standard errors in parentheses. Significance codes: ***: < .001, **: < .01, *: < .05. DV = dependent variable; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion.

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## Web Appendix D: Survey Among TikTok Users

We recruited 40 US participants from the platform Prolific (pre-screened for regular TikTok users; $M_{age}$ = 25.55, SD = 3.54; 28% female) to participate in a survey asking for their familiarity with dance trends on TikTok.

To set up the survey, we compiled a list of 15 current dance trends in the US based on the TikTok "What's trending now" suggestions in the TikTok Creative Center. For each trend, we extracted the number of posted videos for each song on TikTok's song search function at the time the survey was launched (May 2025). This manual information extraction was necessary because TikTok's Creative Center does not directly provide information about the number of videos for each song. Note that this song search information is a conservative estimate because not all users who are using a song are tagging the song, which means that there are likely more videos for each music snippet on TikTok. For each trend, we then extracted three exemplary dance videos from TikTok closely imitating the trend.

In the survey, participants watched the three videos for each trend and then indicated whether they knew the trend ("Have you seen this dance trend before? - Yes / No"). Each participant rated each of the fifteen dance trends, resulting in a total of 600 participant responses for the whole sample. After completing the ratings, participants answered control questions (whether they had technical issues when watching the video, how often they are online on TikTok in minutes per day, how often they watch dance videos on TikTok on a scale from 1 (never) to 7 (very often-almost daily)), and provided their demographic data.

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***Results and Discussion***

Descriptively, we can observe that most trends are known by the majority of TikTok users. For only three trends did fewer than 50% of participants indicate knowing the trend. For nine out of the fifteen trends, more than 60% indicated knowing the trend, and for the three largest trends, 80% or more participants indicated knowing the trend (Table W20).

To test the relationship between trend size and likelihood to know a trend, we calculate a logistic regression on the trend level with the number of trend videos as the predictor and likelihood to know the trend as the dependent variable (1 = yes, 0 = no). Because the number of videos in each trend varies heavily (max = 4,400,000, min = 61,100), we scale the number of videos per trend. We find a significant positive relationship between the number of trend videos and the likelihood of knowing the dance trend ($\beta = .477$, $p < .001$). The results hold when we include trend- or participant-fixed effects (model with trend-fixed effects: $\beta = .499$, $p = .002$; model with participant-fixed effects: $\beta = .750$, $p < .001$). We also find the same results when using raw values instead (all $p \leq .002$).

We derive two important insights from this survey. First, many social media users are familiar with current dance trends on TikTok. The high percentage of users who report being familiar with the trends is noteworthy, particularly when considering that we did not filter for users who regularly watch dance videos, but only for users who indicated that they "ever watch content on TikTok." Second, the results imply that users are more likely to know a trend the more trending videos exist, and speak to the validity of our moderator from the trend exposure interaction.

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Table W20: Overview of Trends Included in the Survey

| Trend ID | Trend Name | Number of Videos Posted | % of Participants Knowing the Trend |
|---|---|---|---|
| **#1** | My Bubble Gum | 325,300 | 72.50 |
| **#2** | In Your Eyes | 381,100 | 62.50 |
| **#3** | Bricks | 62,700 | 62.50 |
| **#4** | Your way's better | 579,400 | 80.00 |
| **#5** | Gimme the light | 61,100 | 52.50 |
| **#6** | Passo Bem Solto | 1,700,000 | 57.50 |
| **#7** | Mona Lisa | 490,400 | 45.00 |
| **#8** | Anxiety | 4,400,000 | 90.00 |
| **#9** | Done with Your Ex | 2,500,000 | 80.00 |
| **#10** | Le Freak | 471,800 | 65.00 |
| **#11** | Heavy Love | 259,300 | 50.00 |
| **#12** | Level Up | 124,800 | 70.00 |
| **#13** | Shake it to the Max | 1,300,000 | 80.00 |
| **#14** | Abusadora | 594,600 | 35.00 |
| **#15** | Super | 202,500 | 42.50 |

*Notes:* Trends were extracted from the "What's trending now" category restricted to the US region in the TikTok Creative Center. The number of videos posted was extracted from TikTok's song search function at the time the survey was launched (May 20, 2025). The percentage of participants knowing the trend was based on survey responses from study participants.

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## Web Appendix E: Experimental Evidence

***Pre-test***

We conducted a pretest to ensure that the videos used in studies 2 and 3 did not vary in terms of video or dance quality. Sixty-two participants ($M_{age}$ = 37.35, SD = 13.32; 51.6% female) were recruited via Prolific and randomly assigned to one of two conditions (atypical vs. typical video). We prescreened for participants who were able to watch videos in their browser.

As an introduction to the survey, participants were informed that the survey was about their perceptions of a social media video. Each participant saw the focal video (either the atypical or the typical video) used in the experiments, which were created by a professional content creator whom we hired to produce the videos exclusively for this research. Then, participants rated the overall video quality ("How do you estimate the overall quality of this video:") and dance quality ("How do you evaluate the quality of the dance (i.e., movements, fit to music, aesthetics, etc.) in this video?") on 7-point Likert scales (1 = very low quality, 7 = very high quality). Finally, participants indicated their demographic data (age, gender, residence, Prolific ID).

Two-sample t-tests reveal no significant differences for overall quality ($M_{atypical}$ = 5.48, SD = 1.43; $M_{typical}$ = 5.10, SD = 1.78, $p$ = .35) or dance quality ($M_{atypical}$ = 4.58, SD = 1.78; $M_{typical}$ = 4.29, SD = 1.55, $p$ = .50). These results indicate that the focal videos did not differ in terms of quality.

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***Additional Details for Study 2***

As an introduction to the study, participants read the following information:

> **This survey: Dancing Trends Videos on TikTok**
>
> TikTok users often dance to trends. Some users are imitating the same choreography that thousands of other users danced. Others use the same music but invent an own dance.
> In this survey, please imagine you are scrolling through your TikTok feed seeing such videos. When watching the videos in this survey, note, that most of them use the same choreography and pay close attention to the dancing. Watch all videos from the beginning to the end and make sure your sound is activated.

As all videos were shown on the same survey page, participants could go back and forth and re-watch videos if they wanted to. Continuing the survey was only possible when participants had watched all the videos. Figure W7 depicts screenshots of the two versions of focal videos (atypical and typical). Figure W8 depicts screenshots of stimulus videos, which were the same for both conditions. To avoid positioning effects, the order of the videos was fixed in both conditions, and the focal video was shown at position 4.

The pre-screening items were yes/no questions, which read as follows: “Do you ever like or proliferate content on social media, such as Instagram, TikTok, Facebook, etc.?” and “For this survey, it is extremely important that you have your audio and a flash player activated to play videos. Do you have your audio and flash player activated?”

Social media usage was captured using the following item: “How often do you like or comment on content on social media?” (1 = never, 7 = very often). Technical issues while watching the video were indicated by answering the following question: “Did you have technical issues while watching the videos?” (yes / no). Demographics included questions regarding age, gender, country of residence, and Prolific ID.

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Figure W7: Focal Videos in the Typical and Atypical Conditions Used in Studies 2 and 3

Typical Video



Atypical Video

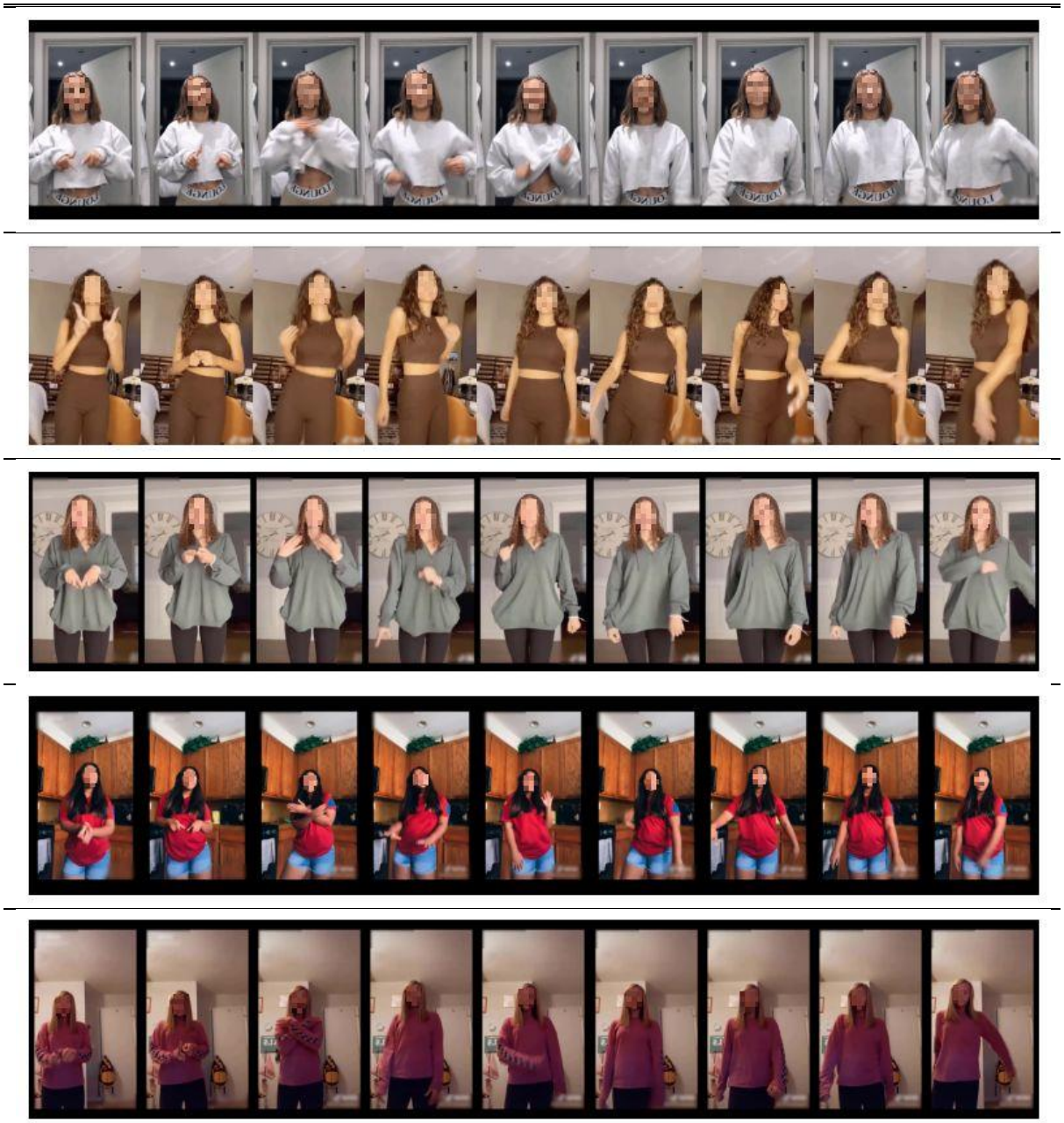


*Notes:* To protect the privacy of the creators, we pixelated their faces.

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Figure W8: Context Videos Used in Study 2 (Same in Both Conditions)



*Notes:* To protect the privacy of the creators, we pixelated their faces.

### *Conceptual Replication of Study 2 with an Additional Atypicality Condition*

For exploratory purposes, we replicated the findings from Study 2. The rationales behind this additional study are the following: First, this study allows us to generalize the findings to another creator and set of videos. Second, it enables us to run an exploratory analysis testing for an additional level of atypicality (moderate atypicality). Again, the focal

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videos kept everything else constant except choreography. A separate pre-test for the focal videos in this study confirmed that the videos did not differ in terms of quality ($N$ = 90 participants; $M_{age}$ = 42.59, SD = 13.33; 52.22% female; ANOVA for overall video quality: $F(2, 89) = 1.32$, $p = .273$; ANOVA for dance quality: $F(2, 89) = 2.09$, $p = .13$).

*Method*

Three hundred participants from Prolific ($M_{age}$ = 42.32, SD = 12.43, 64.33% female, pre-screened for social media users who are able to watch videos in their browser) completed this study, in which they were randomly assigned to one of three conditions: atypical, typical, and moderately atypical video (video screenshots in Figure W9). The manipulation check for atypicality confirmed that both typical videos were seen as more closely adhering to the trend than the atypical video ($M_{atypical}$ = 4.07, SD = 2.11; $M_{typical}$ = 5.33, SD = 1.89; $M_{moderately\ atypical}$ = 4.51, SD = 2.10; typical versus atypical: $t(201) = 4.48$, $p < .001$; typical vs. moderately atypical: $t(196) = 2.90$, $p = .004$).

The study followed a similar design and used similar measures as Study 2, except for the following changes: First, we used a different creator (Figure W9) and different context videos (Figure W10). Second, we did not check for technical issues in the post-questionnaire. Third, we shortened the introduction part of the survey to avoid prompting participants to concentrate on choreography:

**This survey: Dancing Trends Videos on TikTok**

TikTok users often dance to trends. When following trends, users use the same music and perform dances to this music.

In this survey, please imagine you are scrolling through your TikTok feed seeing such dance videos. Watch all videos from the beginning to the end and make sure your sound is activated.

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Figure W9: Focal Videos in the Atypical, Typical, and Moderately Atypical Condition

Atypical Video



Typical Video



Moderately Atypical Video



*Notes:* To protect the privacy of the creators, we pixelated their faces.

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Figure W10: Context Videos for Study 2 Replication (Same in All Conditions)



*Notes:* To protect the privacy of the creators, we pixelated their faces.

*Results*

A chi-square test reveals a significant difference on selecting the focal video between conditions ($\chi^2(1, N = 300) = 7.63$, $p = .022$, *Cramér's V* = .160). For a subsequent logistic regression, we operationalize engagement as the choice of the focal video (1 = yes, 0 = no), and condition as a three-level dummy (0 = typical, 1 = atypical, 2 = moderately atypical).

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The logistic regression confirms that participants are significantly more likely to want to engage with the focal video when it was atypical (37.25%) versus typical (19.80%; $\beta$ = .877, $p$ = .007). The comparison between moderate atypicality (30.93%) and typical video reaches marginal significance ($\beta$ = .595, $p$ = .073). When excluding the typical condition, no significant difference emerges between the atypicality and moderate atypicality conditions ($\beta$ = -.282, $p$ = .347). Accordingly, preferences in the moderately atypical and atypical conditions do not differ, but both atypicality conditions reach—at least directionally—higher engagement.

This finding speaks to our assumption that atypicality, even when it is subtle, can influence engagement.

***Additional Details for Study 3***

*Main study*

We used similar measures as in Study 2 (described in the main manuscript), except that we did not check for social media usage in the post-questionnaire. In this study, participants received the following introduction information:

**This survey: Dancing Trends Videos on TikTok**

TikTok users often dance to trends. Some users are imitating the same choreography that thousands of other users danced. Others use other music or the same music and invent an own dance.

In this survey, please imagine you are scrolling through your TikTok feed seeing such videos.
When watching the videos in this survey, note and pay close attention to the dancing choreographies and how these choreographies differ (or not). Watch all videos from the beginning to the end and make sure your sound is activated.

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*Context videos*

Figure W11 and Figure W12 show the stimulus videos for the low and high exposure conditions. The focal videos were the same as in Study 2.

Figure W11: Context Videos for the Low Exposure Condition in Study 3



*Notes:* To protect the privacy of the creators, we pixelated their faces.

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Figure W12: Context Videos for the High Exposure Condition in Study 3



*Notes:* To protect the privacy of the creators, we pixelated their faces.

### *Study 3 Follow-up: Conceptual replication testing potential mechanisms*

To test the mechanism driving the atypicality effect, we ran an exploratory experiment replicating Study 3 ($N$ = 649 participants from Prolific; $M_{age}$ = 43.00, SD = 13.7; 59.17% female). Specifically, we used the same experimental design, pre-screening, procedure,

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The only difference was that this follow-up study aimed to explore the psychological mechanism for selecting the atypical video. To this end, we added measures after participants selected the video to engage with. First, participants indicated in an open text field their reasons why they selected the video. Then, participants indicated their motive to select this video by answering the following six items on 7-point scales: The extent to which their choice was driven by the creator's (1) creativity, (2) authenticity, (3) competence, as well as their perception of the video as (4) unique, (5) more interesting, and (6) less boring than the other videos (**"**To which degree did you decide to select the video because the creator was..." "more creative / more competent / more authentic... than the others" and **"**To which degree did you decide to select the video because the video was..." "more interesting / less boring / more unique... than the others"; 1 = completely agree; 7 = completely disagree).

Replicating the findings from Study 3, a logistic regression predicting focal video selection revealed a significant interaction of atypicality × trend exposure ($\beta$ = .81, $p$ = .024). More participants selected the focal video in the high exposure condition when it was atypical (36.24%) versus typical (22.60%; $p$ = .007). This difference was not significant in the low exposure condition (22.35% selecting focal video in atypical vs. 25.00% selecting the focal video in typical condition; $p$ = .576).

To examine the underlying motives of participants who selected the focal video in the high exposure condition, we ran paired-sample t-tests for all motives for this condition only. These tests revealed that the video's perceived uniqueness was the strongest reported motive ($M$ = 6.20). Specifically, perceived uniqueness was rated significantly more important than perceiving the video less boring ($M$ = 5.09, $p$ < .001), more interesting ($M$ = 5.44, $p$ < .001), and all creator-related attributes (authenticity, creativity, and competence; all $p$ < .05). Perceived creator creativity (M = 5.85) emerged as the second strongest motive, rated significantly higher than all other motives except for video uniqueness (all $p$ < .05).

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To confirm this finding, we ran additional analyses amongst participants in all conditions who selected to engage with the focal video *(N* = 170; descriptives in Table W21). Specifically, we ran separate 2 (atypicality: typical vs. atypical) × 2 (trend exposure: low vs. high) ANOVAs on the items measuring participants' motives for selecting the focal video. These analyses revealed significant interaction effects on perceived video uniqueness ($F(1, 166) = 16.06$, $p < .001$), and perceived creator creativity ($F(1, 166) = 13.75$, $p < .001$), as well as a marginal interaction effect on perceived creator authenticity ($F(1, 166) = 3.61$, $p = .059$). These motives were consistently considered most important in the high exposure and atypical condition. The interactions were not significant for all other motives (all $p > .12$). Independent-sample t-tests comparing the atypical versus typical conditions within the high-exposure condition confirmed this pattern. The video's perceived uniqueness was a more important reason if the video was atypical versus typical *(*$p < .001$*),* suggesting that participants' motives were strongly driven by uniqueness perceptions in this condition. Similarly, perceiving the creator more creative and authentic was also rated more important in the atypical (vs. typical) condition (both $p < .05$). There was no difference in the tendency to find the atypical video less boring or for all other motives (all $p > .18$).

Overall, these results indicate that the main reason that participants select an atypical video in the high exposure condition, is that they appreciate the atypical videos' uniqueness and the creator's creativity, rather than being bored by the typical videos.

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Table W21: Means and Standard Deviations for Groups

| | **Low Exposure** | | **High Exposure** | |
|---|---|---|---|---|
| | Low Atypicality | High Atypicality | Low Atypicality | High Atypicality |
| Video Interesting | 5.28 (1.63) | 5.10 (1.68) | 5.03 (1.53) | 5.44 (1.49) |
| Video Less Boring | 5.28 (1.60) | 4.78 (1.89) | 4.78 (1.78) | 5.09 (1.50) |
| Video Unique | 5.33 (1.74) | 4.95 (1.88) | 4.58 (1.80) | 6.20 (1.11) |
| Creator Creative | 4.89 (1.77) | 5.03 (1.94) | 3.85 (1.75) | 5.85 (1.07) |
| Creator Competent | 5.11 (1.69) | 5.20 (1.76) | 4.75 (1.90) | 4.96 (1.49) |
| Creator Authentic | 5.50 (1.76) | 5.35 (1.61) | 4.80 (1.92) | 5.61 (1.28) |
| N | 36 | 40 | 40 | 54 |

*Notes*: The statistics only show those who selected the focal video. Standard deviation in parentheses. For all items we used "more interesting / unique / creative / authentic", but we asked for "less boring".

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## Web Appendix F: Additional Information on Field Experiment

To compile a list of dance trends in the US, we searched TikTok's Creative Center for "Song Trends" and filtered for US trends in the last 30 days (as of May 2025). We then selected the five most popular dance trends:

- Heavy Love
- Done with Your Ex
- Passo Bem Solto
- Shake it to the Max
- Abusadora

Table W22 shows an overview of sample statistics for the field experiment.

Table W22: Overview of Sample Statistics

| | **Mean** | **Median** | **SD** | **Min** | **Max** |
|---|---|---|---|---|---|
| Creator Number of Videos | 541.81 | 217 | 832.23 | 14 | 4,141 |
| Creator Number of Followers | 41,754.14 | 3,860 | 158,018 | 10 | 949,280 |
| Number of Views | 783.26 | 434 | 1,039.12 | 20 | 5,375 |
| Number of Likes | 58.64 | 39 | 64.14 | 1 | 333 |
| Number of Days Online | 32.93 | 36 | 10.25 | 17 | 48 |

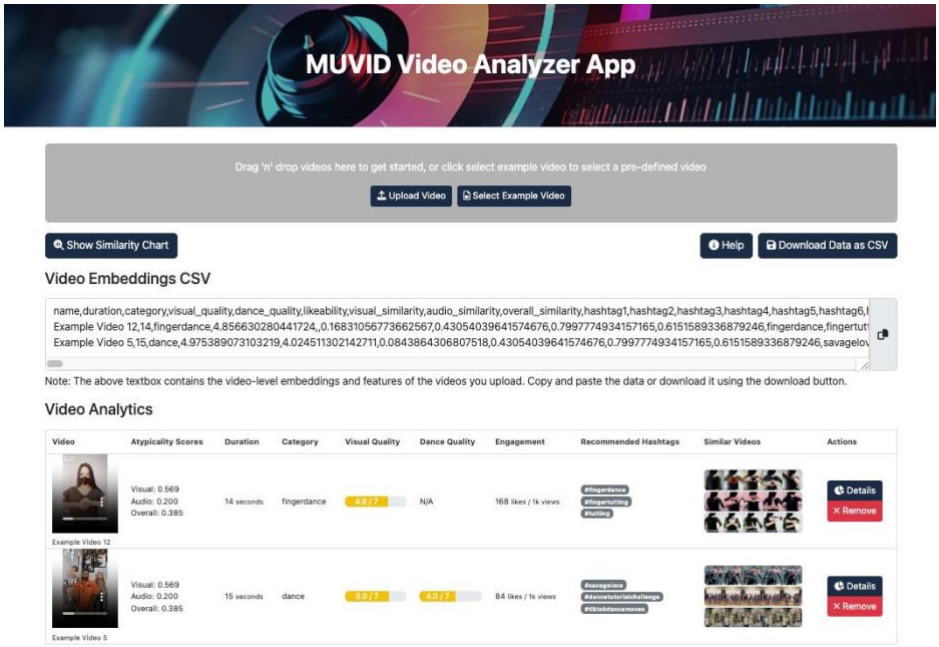
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# Web Appendix G: Research App

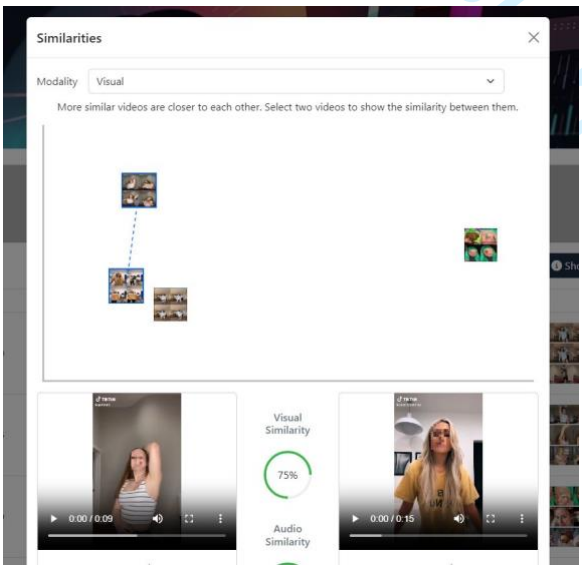
## *Screenshots of the MUVID App Interface*

Figure W13: Screenshot Showing the Main Page of the MUVID App




Notes: Two videos that were processed using the MUVID app. The app computes the video embeddings, which can be downloaded as CSV, estimates atypicality scores, predicts the category, visual quality, and engagement, recommends suitable hashtags, and shows similar videos. To protect the privacy of the creators, we pixelated their faces.

Figure W14: Screenshot of the MUVID App Showing More Similar Videos Closer to Each Other

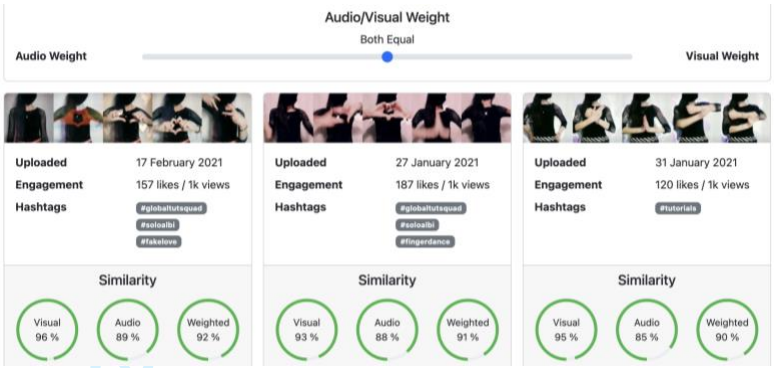



*Notes*: The uploaded videos are displayed for the selected modality in a 2D space after applying a PCA transformation. The app allows for

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comparing two selected videos by showing their analytics side by side and computing their similarities in the visual and audio domains. To protect the privacy of the creators, we pixelated their faces.

Figure W15: Screenshot of the MUVID App Showing Similar Videos for a Given Focal Video




*Notes:* For a given focal video, the most similar videos from a dataset of 100,000 are shown. The slider above allows the user to modify the weightings of the audio or visual domain.

### *Implementation Details*

The MUVID app's backend is implemented using Python 3.10, where we used the FastAPI web framework to build an API that is called from the frontend. The backend processes the uploaded videos by extracting visual and audio embedding features, which are then used as input for all prediction models. We do not save the videos on our server, but instead, store the embeddings and analytics within the user's browser session. Whenever details for a specific video are required again, the embeddings stored on the browser are sent to the server. Once the user closes the browser, all data is deleted.

The frontend of the MUVID app was implemented using VueJS, a popular JavaScript framework known for its modular and component-based approach to frontend development. It communicates with the backend API through HTTP requests, retrieving data and sending user inputs for processing. To facilitate development and deployment, we organize the frontend and backend into two containers using Docker.

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***Content Characteristics Prediction Models***

We applied transfer learning to train five models, which are already integrated within the app. These models predict relevant content characteristics of the videos and can help determine which videos will have more impact. The predicted scores of these content characteristics are calculated for uploaded videos and can easily be exported.

*Engagement prediction*

To develop an accurate engagement prediction model, we utilize the engagement dataset obtained during the downstream evaluation phase. We replace the linear regression model from the downstream evaluation with a multi-layer perceptron comprising three hidden layers. Finally, the model is trained using the MUVID multimodal embeddings (i.e., the combination of visual and audio embeddings), and subsequently evaluated on a test set using an 80%/20% split. The model achieves an $R^2$ score of .307 and an MAE of .489. To enhance the interpretability of the engagement predictions, we normalize the predicted values to derive a measure of likes per 1,000 views.

*Category prediction*

To train an accurate category classifier, we reused the labeled category dataset (consisting of 14 different categories) constructed in the downstream evaluation and replaced the linear model with a multi-layer perceptron that consists of three hidden layers. Training the model using our multimodal embeddings resulted in an accuracy of 54.61% measured on an unseen test set (80%/20% split).

*Quality prediction*

We integrated the visual quality and dance quality regressors developed in the result section to predict the quality of the uploaded videos and the dance in particular.

*Hashtag recommender*

To build the hashtag recommender, we identify the top 1,000 hashtags in the dataset

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and generate hashtag embeddings by computing the average embedding of all videos associated with a specific hashtag. Then, to recommend suitable hashtags, we compute the cosine similarity of the videos to all hashtags and return the top hashtags.

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## References


Akpinar, Ezgi and Jonah Berger (2017), "Valuable virality," *Journal of Marketing Research*, 54 (2), 318–30.

Al-Tahan, Haider and Yalda Mohsenzadeh (2021), "CLAR: Contrastive Learning of Auditory Representations," in *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics*, Proceedings of Machine Learning Research, Arindam Banerjee and Kenji Fukumizu, eds., PMLR, 2530–38.

Baverstock, Alex (2019), "A New Type of Data: Why Video Research Has Ballooned," *ipsos*, (accessed December 27, 2022), https://www.ipsos.com/en-uk/new-type-data-why-video-research-has-ballooned.

Becker, Jan-Michael, Dorian Proksch, and Christian M. Ringle (2022), "Revisiting Gaussian copulas to handle endogenous regressors," *Journal of the Academy of Marketing Science*, 50 (1), 46–66.

Berger, Jonah, Ashlee Humphreys, Stephan Ludwig, Wendy W. Moe, Oded Netzer, and David A. Schweidel (2020), "Uniting the Tribes: Using Text for Marketing Insight," *Journal of Marketing*, 84 (1), 1–25

Bhandari, Aparajita and Sara Bimo (2020), "TikTok and the 'Algorithmized Self': A New Model Of Online Interaction," *Spirales*.

Boichuk, Jeffrey P., Raghu Bommaraju, Michael Ahearne, Florian Kraus, and Thomas J. Steenburgh (2019), "Managing laggards: The importance of a deep sales bench," *Journal of Marketing Research*, 56 (4), 652–665.

boyd, danah (2010), "Social Network Sites as Networked Publics: Affordances, Dynamics, and Implications," *A networked self*, 47–66.

Business Insider (2017), "Video will account for an overwhelming majority of internet traffic by 2021," *Business Insider*, (accessed December 22, 2022), https://www.businessinsider.com/heres-how-much-ip-traffic-will-be-video-by-2021-2017-6.

Campbell, Colin and Justine Rapp Farrell (2020), "More than meets the eye: The functional components underlying influencer marketing," *Business Horizons*, 63 (4), 469–79.

Cascio Rizzo, Giovanni Luca, Francisco Villarroel Ordenes, Rumen Pozharliev, Matteo De Angelis, and Michele Costabile (2024), "How high-arousal language shapes micro- versus macro-influencers' impact," *Journal of Marketing*, 88 (4), 107–128.

Ceci, Laura (2022), "Most popular content categories on TikTok worldwide as of July 2020, by number of hashtag views," *Statista*, (accessed April 16, 2023), https://www.statista.com/statistics/1130988/most-popular-categories-tiktok-worldwide-hashtag-views/.

Chen, Ting, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton (2020), "A Simple Framework for Contrastive Learning of Visual Representations," PMLR.

Eggert, Andreas, Lena Steinhoff, and Carina Witte (2019), "Gift purchases as catalysts for strengthening customer–brand relationships," *Journal of Marketing*, 83 (5), 115–132.

Freer, Anne (2022), "These are the top 10 hashtag categories on TikTok," *Business of Apps*, (accessed October 7, 2022), https://www.businessofapps.com/news/these-are-the-top-10-hashtag-categories-on-tiktok/.

Garnefeld, Ina, Andreas Eggert, Sabrina V. Helm, and Stephen S. Tax (2013), "Growing existing customers' revenue streams through customer referral programs," *Journal of Marketing*, 77 (4), 17–32.

Gong, Yuan, Yu-An Chung, and James Glass (2021), "AST: Audio Spectrogram Transformer," *arXiv [cs.SD]*.


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Author Accepted Manuscript

Graffius, Scott (2025), "Lifespan (Half-Life) of Social Media Posts: Update for 2025," *Scott Graffius*, (accessed July 23, 2025), https://www.scottgraffius.com/blog/files/lifespan-halflife-of-social-media-posts-update-for-2025.html.

Han, Nuo, Sijia Li, Feng Huang, Yeye Wen, Yue Su, Linyan Li, Xiaoqian Liu, and Tingshao Zhu (2023), "How social media expression can reveal personality," *Frontiers in Psychiatry*, 14, 1052844.

Hartmann, Jochen, Mark Heitmann, Christina Schamp, and Oded Netzer (2021), "The power of brand selfies," *Journal of Marketing Research*, 58 (6), 1159–1177.

He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun (2016), "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, openaccess.thecvf.com, 770–78.

Howard, Andrew G., Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam (2017), "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," *arXiv [cs.CV]*.

Kay, Will, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, Mustafa Suleyman, and Andrew Zisserman (2017), "The Kinetics Human Action Video Dataset," *arXiv [cs.CV]*.

Kern, Margaret L., Gregory Park, Johannes C. Eichstaedt, H. Andrew Schwartz, Maarten Sap, Laura K. Smith, and Lyle H. Ungar (2016), "Gaining insights from social media language: Methodologies and challenges," *Psychological Methods*, 21 (4), 507–25.

Klug, Daniel, Yiluo Qin, Morgan Evans, and Geoff Kaufman (2021), "Trick and Please. A Mixed-Method Study On User Assumptions About the TikTok Algorithm," in *Proceedings of the 13th ACM Web Science Conference 2021*, WebSci '21, New York, NY, USA: Association for Computing Machinery, 84–92.

Le-Khac, Phuc H., Graham Healy, and Alan F. Smeaton (2020), "Contrastive Representation Learning: A Framework and Review," *IEEE Access*, 8, 193907–34.

Li, Xi, Mengze Shi, and Xin Wang (2019), "Video mining: Measuring visual information using automatic methods," *International Journal of Research in Marketing*, 36 (2), 216–31.

Lie, David S., Ali Gohary, Pei-Yu Chien, and Bach T. N. Truong (2024), "Why do you find similar reviews helpful? Psychological mechanisms of the effect of linguistic style matching on review helpfulness," *Decision Support Systems, 187,* 114340.

Liengaard, Benjamin D., Jan-Michael Becker, Mikkel Bennedsen, Phillip Heiler, Luke N. Taylor, and Christian M. Ringle (2025), "Dealing with regression models' endogeneity by means of an adjusted estimator for the Gaussian copula approach," *Journal of the Academy of Marketing Science*, 53 (1), 279–299.

Liu, Jia and Ziwei Cong (2023), "The Daily Me Versus the Daily Others: How Do Recommendation Algorithms Change User Interests? Evidence from a Knowledge-Sharing Platform," *Journal of Marketing Research*, 60 (4), 767–91.

Lu, Shasha, Li Xiao, and Min Ding (2016), "A Video-Based Automated Recommender (VAR) System for Garments," *Marketing Science*, 35 (3), 484–510.

Mahajan, Rhea, Remia Mahajan, Eishita Sharma, and Vibhakar Mansotra (2022), "'Are we tweeting our real selves?' personality prediction of Indian Twitter users using deep learning ensemble model," *Computers in Human Behavior*, 128, 107101.

Park, Daniel S., William Chan, Yu Zhang, Chung-Cheng Chiu, Barret Zoph, Ekin D. Cubuk, and Quoc V. Le (2019), "SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition," *arXiv [eess.AS]*.

Park, Sungho and Sachin Gupta (2012), "Handling endogenous regressors by joint estimation using copulas," *Marketing Science*, 31 (4), 567-586

Author Accepted Manuscript

Pennebaker, James W., Martha E. Francis, and Roger J. Booth (2001), "Linguistic Inquiry and Word Count: A computerized text analysis program."

Pennebaker, James W. and Laura A. King (1999), "Linguistic styles: Language use as an individual difference," *Journal of personality and social psychology*, 77 (6), 1296–1312.

Qian, Rui, Tianjian Meng, Boqing Gong, Ming-Hsuan Yang, Huisheng Wang, Serge Belongie, and Yin Cui (2021), "Spatiotemporal contrastive video representation learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, openaccess.thecvf.com, 6964–74.

Radford, Alec, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever (2021), "Learning Transferable Visual Models From Natural Language Supervision."

Rajaram, Prashant and Puneet Manchanda (2023), "Influencer Videos: Unboxing the Mystique," *Available at SSRN 3752107.*

Reimers, Nils and Iryna Gurevych (2019), "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," *arXiv [cs.CL]*.

Rosenbaum, Paul R., and Donald B. Rubin (1983), "The central role of the propensity score in observational studies for causal effects," *Biometrika*, 70 (1), 41–55.

Schiappa, Madeline C., Yogesh S. Rawat, and Mubarak Shah (2023), "Self-Supervised Learning for Videos: A Survey," *ACM Computing Surveys*, 55 (13s), 1–37.

Schwenzow, Jasper, Jochen Hartmann, Amos Schikowsky, and Mark Heitmann (2021), "Understanding videos at scale: How to extract insights for business research," *Journal of Business Research*, 123, 367–79.

Simpson, Ellen and Bryan Semaan (2021), "For You, or For"You"? Everyday LGBTQ+ Encounters with TikTok," *Proc. ACM Hum.-Comput. Interact.*, 4 (CSCW3), 1–34.

Smith, Andrew N., Eileen Fischer, and Chen Yongjian (2012), "How Does Brand-related User-generated Content Differ across YouTube, Facebook, and Twitter?," *Journal of Interactive Marketing*, 26 (2), 102–13.

Sohn, Kihyuk (2016), "Improved deep metric learning with multi-class n-pair loss objective," *Advances in Neural Information Processing Systems*.

Teixeira, Thales, Rosalind Picard, and Rana el Kaliouby (2014), "Why, When, and How Much to Entertain Consumers in Advertisements? A Web-Based Facial Tracking Field Study," *Marketing Science*, 33 (6), 809–27.

Teixeira, Thales S., Michel Wedel, and Rik Pieters (2010), "Moment-to-Moment Optimal Branding in TV Commercials: Preventing Avoidance by Pulsing," *Marketing Science*, 29 (5), 783–804.

Teixeira, Thales, Michel Wedel, and Rik Pieters (2012), "Emotion-Induced Engagement in Internet Video Advertisements," *Journal of Marketing Research*, 49 (2), 144–59.

Tian, Zijun, Ryan Dew, and Raghuram Iyengar (2024), "Mega or Micro? Influencer Selection Using Follower Elasticity," *Journal of Marketing Research*, 61 (3), 472–95.

Tong, Zhan, Yibing Song, Jue Wang, and Limin Wang (2022), "VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training," *Advances in Neural Information Processing Systems*, 35, 10078–93.

Tran, Du, Heng Wang, Lorenzo Torresani, Jamie Ray, Yann LeCun, and Manohar Paluri (2018), "A closer look at spatiotemporal convolutions for action recognition," In *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition* (pp. 6450-6459).

Peer Review Version

Author Accepted Manuscript

Warren, Nooshin L., Matthew Farmer, Tianyu Gu, and Caleb Warren (2021), “Marketing Ideas: How to Write Research Articles that Readers Understand and Cite,” *Journal of Marketing*, 85 (5), 42–57.

Wlömert, Nils, and Dominik Papies (2019), "International heterogeneity in the associations of new business models and broadband internet with music revenue and piracy," *International Journal of Research in Marketing*, 36 (3), 400–419.

Yang, Jeremy, Juanjuan Zhang, and Yuhan Zhang (2025), “Engagement that sells: Influencer video advertising on TikTok,” *Marketing Science*, 44 (2), 247-267.

Zhou, Mi, George H. Chen, Pedro Ferreira, and Michael D. Smith (2021), “Consumer Behavior in the Online Classroom: Using Video Analytics and Machine Learning to Understand the Consumption of Video Courseware,” *Journal of Marketing Research*, 58 (6), 1079–1100.

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